

The Qualia Tomogram:

A Generalized Projection-Slice Framework for Receiver-Conditioned Neural Reconstruction

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Abstract

A brain does not receive one finished description of an object, body, memory, or intended action. Different cells, arrays, modalities, and moments supply partial measurements whose effects depend on learned receiver structure and current state. The scientific problem is whether relations among those partial views support a shared, body-relative reconstruction that predicts an unseen view and changes selectively when one declared relation is perturbed. This paper calls such a candidate reconstruction a **SAN Tomogram** only after the operation and its failure conditions are specified. The broader framework, Tomographic Coherence-Based Neural Rendering, combines receiver-conditioned transforms, tonic context, informative phase-wave differentials, recurrent projection, action-dependent sampling, and returned evidence without positing an inner viewer.

The biological program is prospective. It defines a locked analysis of public macaque reach-to-grasp recordings, comparing structured phase relations with evoked activity, firing rate, amplitude, power, scalar coherence, temporal dynamics, raw-signal models, and capacity-matched nonlinear alternatives. The primary endpoint is held-out prediction of the cue-instructed grip condition. Because cue identity and grip instruction are confounded in the source task, the protocol cannot establish a maintained motor plan independent of the cue and reports no neural or consciousness result.

Two deterministic synthetic reference applications establish executability and expose failure modes. In the primary generator, the full typed inverse reconstructs the latent state with RMSE 0.013853 and predicts a withheld receiver with RMSE 0.031271. It beats its mechanistic ablations, while raw linear ridge remains a near-tied and slightly better predictor on average across six smaller replications. Receiver deletion, phase scrambling, exact rescue, nonspecific rescue, carrier drift, targeted writing, and overdrive are tested. Phase-relational features succeed only in the phase-bearing generator; amplitude geometry is strongest in the adverse generator. Ten application tests and ten executable-mathematics tests pass, and the governed generated artifacts reproduce byte-for-byte.

The mathematical layer separates uniqueness, stability, local and global identifiability, restricted carrier invariance, and semantic anchoring. It supplies a rank-deficient counterexample, latent-coordinate ambiguity, and explicit intervention algebra. An isolated Lean 4.26 kernel machine-checks fifteen bounded theorems with build and axiom-audit exit code zero and no standalone proof placeholders. Those theorems establish exact consequences of declared abstract premises; they do not prove that neural tissue realizes the model, that a latent coordinate is a qualia, or that a biological write can isolate a coordinate. A twenty-item steelman-first audit preserves these negative and open results. The contribution is therefore a falsifiable, application-backed theory-and-methods program with an explicit route from synthetic sufficiency to neural prediction, causal specificity, and eventual phenomenal discrimination—not a completed measurement of conscious experience.

Part I — The Problem, Established Science, And A Bounded Neural Test

1 Claim, Scope, And The Empirical Firewall

A person looking at broccoli does not receive a finished broccoli token. Different receptors and circuits register color, contour, depth, smell, taste, touch, body position, expectation, and possible action. The unexplained problem is not whether any one of these variables can be decoded. It is how the nervous system preserves their differences while constructing a body-relative state that predicts what another receiver will register and what an action will change.

Established neuroscience supplies important parts of this problem. Neural population dynamics can carry task structure that is not reducible to one neuron [44]. Sequential latent-state models can infer single-trial population trajectories [45], and jointly learned embeddings can align neural activity with behavior under declared supervision [46]. Predictive-coding models formalize top-down predictions and bottom-up residuals [47]. Multisensory neurons can integrate visual, auditory, and somatosensory inputs [53]. None of those results alone identifies one biological forward operator, proves a unique latent scene, turns a decoder into a stimulator, or establishes phenomenal content.

The plain-language proposal is therefore this. Treat each measured cell, array, modality, or moment as a partial view whose transformation depends on the receiver’s learned structure and current condition. Use several views to estimate a shared changing state. Project that estimate back through the declared receiver models. Test it on views withheld from fitting. Then disturb one relation and ask whether the predicted component fails; restore that relation under matched nonspecific drive and ask whether the same component returns. Multiview theory shows that joint views can improve identifiability only under explicit assumptions; arbitrary nonlinear mixing remains underdetermined without them [60,61].

Only after that operation is clear do the names become useful. A consequential receiver-relative difference is a Phase-Wave Differential (PWD). The receive-transform-project-recur architecture is Neural Array Projection Oscillation Tomography (NAPOT). The active distributed reconstruction is a SAN Tomogram. Tomographic Coherence-Based Neural Rendering (TCNR) names the wider theory-and-methods program. The words do not supply evidence; held-out prediction, perturbation, rescue, nuisance control, and failure conditions do.

In ordinary language, the central question is whether the *arrangement* of activity across the array helps identify the instructed condition after the best available descriptions of the activity itself have already been supplied. Existing neuroscience provides signal processing, decoding, traveling-wave analysis, communication-through-coherence hypotheses, latent-state models, and causal perturbation methods. TCNR adds a particular joined operation: partial measurements are related in space and time, the relation is used to estimate a task-relevant latent state, that estimate is compared with equal-input alternatives, and a later perturbation asks whether the relation changes receiver transfer. “Tomographic” is therefore earned only when a forward operator and reconstruction test are identified; otherwise the narrower phrase is *relational recurrent reconstruction*.

TCNR names the full theory-and-methods program. The frozen empirical model label **M6-TNR** is retained from the preregistration and denotes the narrower tomographic neural-rendering feature model tested in Stage A; it should not be read as the complete TCNR program.

The Stage A claim is:

During the instructed delay, engineered beta-band spatial phase features improve out-of-sample

prediction of the cue-instructed precision-versus-side grip condition under the declared representation classes, relative to evoked activity, spike/rate summaries, amplitude, power, scalar coherence, temporal history, an unengineered 1-kHz LFP comparator, and an equal-compute nonlinear LFP comparator.

The target is an experimentally defined cue-instructed task condition, not a private percept. The first fixed LED cue directly specifies PG versus SG in this dataset, so cue identity and instructed grip are not separately identifiable. Evoked-response controls can test whether a declared feature class improves prediction beyond a cue-locked waveform comparator; they cannot make the motor-plan interpretation independent of cue identity. A positive result would support only one bounded TNR component: engineered phase structure has useful held-out predictive bias under the declared model classes. It would not establish new information absent from the LFP, phase necessity, a literal tomographic inversion, a cue-independent maintained plan, or consciousness.

The protocol separates three questions:

1. **Representation utility:** do phase-geometry features improve held-out prediction under the declared comparator classes?
1. **Mechanistic specificity:** does the improvement survive controls that preserve power, autocorrelation, common drive, or raw input?
1. **Causation:** would selective phase perturbation change receiver transfer at matched energy?

Stage A addresses the first two observationally. Stage B states a prospective intervention and is not represented as completed.

The conceptual layer asks a different question: what structure would connect difference detection, latent reconstruction, action selection, and consciousness-relevant access without confusing a metaphor with a measurement? It may motivate variables, alternatives, and future experiments, but it cannot alter the frozen Stage A target, folds, comparator ladder, SESOI, or decision rule after outcomes are inspected. No philosophical premise is allowed to function as evidence for a neural effect, and no neural decoding result is allowed to function as proof of a metaphysical claim.

2 Minimal Reconstruction Model

Let C_i be the cue-instructed grip condition on trial i , with values precision grip (PG) or side grip (SG). Let Y_i denote the multichannel neural observation in the pre-GO delay window. A task-relative render is the calibrated out-of-fold probability

$$q_i = P(C_i = PG | Z_i),$$

where Z_i is a declared feature set. The word *render* means a task-relative estimate available to a decoder or downstream controller. It does not mean a phenomenal image.

For electrode j , beta-band analytic activity is

$$x_j(t) = a_j(t) \exp(i\phi_j(t)), \phi_j(t) = \arg x_j^{(13-30Hz)}(t).$$

For undirected neighboring electrodes (j,k) , define the wrapped phase difference

$$\Delta \phi_{jk}(t) = \arg \exp(i[\phi_k(t) - \phi_j(t)]).$$

This quantity does not by itself assign causal direction. At each time point, fit the wrapped spatial plane

$$\phi_j(t) \approx \alpha_t + k_x(t) x_j + k_y(t) y_j \pmod{2\pi}.$$

The fitted gradient is $\mathbf{g}_t = (k_x(t), k_y(t))$. A continuous lawful-gradient score is

$$L_t = |(1/N) \sum_j \exp(i[\phi_j(t) - \alpha_t - \mathbf{g}_t \cdot \mathbf{r}_j])|.$$

TNR features include gradient magnitude and direction, circular residual, local curl/divergence-like summaries on the array graph, temporal stability, and nearest-neighbor phase differences. No confirmatory conclusion depends on a binary wave/non-wave threshold.

3 Frozen Dataset And Task

3.1 Immutable data route

The source dataset is Brochier et al., *Massively parallel recordings in macaque motor cortex during an instructed delayed reach-to-grasp task*:

- data DOI: 10.12751/g-node.f83565;
- data repository: https://gin.g-node.org/INT/multielectrode_grasp;
- frozen repository commit: a6d508be099c41b4047778bc2de55ac216f4e673;
- article DOI: 10.1038/sdata.2018.55;
- data license: CC BY 4.0;
- repository code license: BSD 3-Clause.

The repository's tag 1.1.0 resolves to commit 24cd5cae3ae79066ca37844cab931d04dcad977, but that tagged tree does **not** contain the later datasets_nix exports. This protocol therefore freezes the exact later commit above rather than combining the tag identity with files it does not contain.

The reduced 1-kHz NIX files at the frozen commit are:

Subject	File	Annex identity
monkey L	datasets_nix/1101210-001_no_raw.nix	MD5 f20162beefd2af1077325c445035a9d0; 882,371,719 bytes
monkey N	datasets_nix/1140703-001_no_raw.nix	MD5 5c5e443f3b53361a1bed3ec1bcad1b0b; 779,109,880 bytes

The files are not included in this manuscript package. Their local download hashes must be checked against the annex identities and recorded before analysis.

3.2 Development and replication roles

- development animal: monkey L (*l101210* – 001);
- locked second-animal robustness subject: monkey N (*i140703* – 001);
- target: binary cue-instructed grip condition, PG versus SG;
- primary window: GO-ON - 900 ms through GO-ON - 100 ms;
- primary band: 13-30 Hz;
- primary unit: one prediction and one loss per trial;
- inclusion: correct trials with complete CUE-OFF and GO-ON events and dataset-QC-passing LFP;
- exclusion decisions are label-blind and recorded before grip-conditioned scoring.

All transforms, feature definitions, baseline selection, model capacity, hyperparameter-selection rules, surrogates, and decision thresholds are developed on monkey L. Monkey N's condition-specific outcomes are not inspected until that configuration is frozen. Monkey N is a locked second-animal robustness test, not a direct anatomical replication: monkey L spans M1/PMd whereas monkey N is more lateral and includes PMv. Array coordinate orientation and hemisphere conventions must be frozen in the executable lock.

4 Signal Preparation And Feature Reliability

4.1 Primary pipeline

1. Load the reduced NIX recording and odML/Neo annotations.
2. Preserve the physical 10-by-10 Utah-array coordinates and four inactive positions.
3. Apply the dataset's channel/trial quality annotations without viewing label-conditioned outcomes.
4. Run the primary estimator causally on each continuous recording in acquisition order; no sample

later than time t may contribute to the phase estimate reported at t .

1. End all confirmatory input 100 ms before GO-ON. The primary pipeline must embargo the estimator's declared latency and filter state so no sample after the cutoff, fold boundary, or trial boundary can enter a confirmatory feature.

1. Estimate 13-30 Hz phase with a strictly past-only state-space oscillator whose transition, observation noise, initialization, latency, missing-sample rule, and cutoff handling are frozen in the executable lock. A centered Butterworth-plus-Hilbert estimate is retrospective sensitivity analysis only and may never support the primary pre-GO claim.

1. Fit all centering, scaling, PCA, feature-selection, offset, and model parameters inside training folds only.

4.2 Sensitivity set

The bounded sensitivity set crosses:

- original reference, common-average reference, and 2-D surface Laplacian;
- the primary strictly causal state-space estimator and retrospective offline-Hilbert sensitivity;
- phase-quality thresholds at the training-fold 10th, 20th, and 30th amplitude percentiles.

The primary threshold is the 20th percentile. The threshold is fit per channel from training-fold samples

only. The resulting time-point validity mask is supplied identically to every comparator, retained samples are reported by condition without outcome-based tuning, and a no-threshold sensitivity is mandatory. The confirmatory conclusion fails the robustness gate if its sign reverses across this bounded set.

5 Comparator Ladder And Capacity Contract

Every model receives the same outer folds, preprocessing access, optimizer budget, and locked capacity-selection rule.

Model	Feature family
M0	intercept and training-fold class prevalence
M1-rate	spike/event rate, broadband and cue-evoked summaries, trial-order nuisance
M2-power	M1 plus sitewise beta amplitude and power
M3-sync	M2 plus scalar PLV/coherence with no spatial gradient
M4-dynamics	non-oscillatory linear state/history features
M5-linear-LFP	unengineered 1-kHz LFP sequence plus geometry and history, compressed within fold and decoded linearly
M5b-nonlinear-LFP	the same unengineered 1-kHz LFP, mask, geometry, and history through an equal-compute causal temporal model capable of learning filtering, pairwise interactions, and temporal structure without supplied phase features
M6-TNR	the same input access plus declared phase-gradient, wave-geometry, and edge-relative phase features

The linear confirmatory family uses elastic-net logistic regression. Training-fold-only PCA chooses q from $\{8,16,24,32\}$ on monkey L. Equal PCA dimension is not treated as equal representational capacity. The nonlinear M5b comparator and M6 receive the same input samples, validity mask, training-example budget, hyperparameter-evaluation count, early-stopping rule, and maximum multiply-accumulate budget. Their exact architectures, initialization seeds, penalty grids, and optimization rules remain release blockers until frozen machine-readably. The selected q , capacity rule, and baseline champion are locked before monkey N.

6 Estimands, Nulls, And Decision Rule

Let $L(M)$ be out-of-fold binary log loss in bits per trial. Let M_{champion} be the strongest non-TNR comparator selected on monkey L under the locked rule. The primary paired increment is

$$\Delta_{\text{TNR}} = L(M_{\text{champion}}) - L(M6 - \text{TNR}).$$

Positive values favor the phase-structured model. The superiority null is

$$H0_{\text{meaningful}}: \Delta_{\text{TNR}} \leq 0.02 \text{ bits/trial}.$$

The smallest effect of interest is 0.02 bits/trial. Meaningful support on monkey N requires:

1. Δ_{TNR} is at least 0.02 bits/trial;
2. a one-sided 95% full-pipeline block-bootstrap lower bound is above 0.02 bits/trial; and
3. the monkey-L development estimate has the same sign.

The equivalence null is

$H_{eq}: |\Delta_{TNR}| \leq 0.02$.

If the 90% confidence interval is entirely inside $[-0.02, +0.02]$, the meaningful representation- utility claim fails for this preparation. Positive estimates whose lower bound is above zero but not above 0.02 are reported as positive-but-sub-SESOI, not meaningful support. The other fixed outcomes are equivalent, inconclusive, or harmful.

Secondary estimands are held-out AUROC, balanced accuracy, next-100-ms LFP Gaussian log score, post-GO reaction-time prediction, and effect stability across the sensitivity set. Error prediction is removed because the primary cohort excludes error trials; any future error endpoint requires a separate all-trials cohort and lock. The log-score increment is not called literal mutual information unless calibration and density conditions are separately established.

7 Null And Surrogate Ensemble

Each null declares what it is intended to preserve and destroy. Those preservation claims must be verified numerically on the realized surrogate before interpretation, and the complete preprocessing, selection, and fitting pipeline is rerun for every replicate.

1. **Label permutation within contiguous temporal quintiles:** preserves slow drift and class counts; destroys label association.
1. **Whole-trial multichannel phase-map shuffle within temporal quintile:** preserves each map's power and spatial organization; destroys trial-label alignment.
1. **Independent per-channel Fourier phase scrambling:** preserves channel amplitude spectra; destroys cross-site phase relations.
1. **Independent circular shifts by preregistered lags longer than two beta cycles:** preserves autocorrelation and power; destroys inter-site and event alignment.
1. **Distance/degree-constrained array-coordinate permutation:** preserves signals and graph density; destroys anatomical embedding.
1. **Edge/preferred-offset shuffle:** preserves phase marginals; destroys the proposed stitching relation.
1. **Cue-locked evoked waveform and raw-sequence champion:** tests whether engineered phase features merely repackage evoked activity already available in the raw signal.
1. **Cycle-shape and acquisition controls:** test nonsinusoidal harmonics, aperiodic slope, mains contamination, sampling, and filter-edge artifacts.

The executable lock must additionally freeze the shuffle unit and strata; FFT window and padding; circular-shift lag distribution; coordinate-permutation group; random seeds; number of replicates; and model-refit rule. Until those are machine-readable, this is a pre-registration specification, not a signed analysis lock. Max-statistic permutation inference controls the surrogate family. Exploratory secondary endpoints use

Benjamini-Hochberg false-discovery-rate correction. There is one confirmatory band, window, target, and primary model contrast.

8 Splits, Uncertainty, And Power

Monkey L uses five contiguous blocked outer folds with blocked nested inner cross-validation. Monkey N uses five contiguous blocked cross-fitting folds with the locked pipeline. Only out-of-fold trial predictions enter scoring. The complete preprocessing, hyperparameter selection, training, and scoring pipeline is rerun inside every bootstrap or permutation replicate; resampling a stored loss sequence alone is forbidden because outer training sets overlap. The primary moving-block length is 20 consecutive eligible trials, with 10- and 40-trial sensitivity analyses. Inference is conditional on these two recordings and is not a population-level macaque estimate.

This is a fixed-size public dataset. Before opening monkey N's grip-conditioned outputs:

1. use monkey L's paired out-of-fold loss sequence;
2. run at least 2,000 moving-block simulations at effects from 0 through 0.10 bits/trial;
3. report estimated power at the fixed 0.02 bits/trial effect; and
4. report the minimum detectable effect at 80% power.

If monkey N cannot supply 80% simulated power at the smallest effect of interest, Stage A remains a feasibility/estimation study regardless of a nominal p-value.

9 Stage A Execution Contract

The analysis is complete only when a clean environment can reproduce:

- a data manifest containing repository commit, file names, annex identities, downloaded hashes, trial counts, and QC exclusions;
- a configuration file with no outcome-selected confirmatory parameter and exact feature algorithms, graph edges, circular/spatial formulas, temporal summaries, model architectures, grids, optimizers, seeds, estimator state/latency rules, surrogate construction, refitting, and bootstrap rules;
- deterministic outer/inner fold assignments;
- out-of-fold predictions for every included trial and model;
- the complete surrogate ensemble;
- the power replay and uncertainty intervals;
- an all-runs manifest including failed and exploratory runs; and
- a machine-readable results file whose scope says `physical_intervention=false` and `consciousness_test=false`.

No confirmatory wording is written from an in-sample fit or from monkey N before the lock file, software environment, and immutable trial/fold assignment tables are signed and hashed. The present protocol still does not claim that the future empirical execution lock is complete.

10 Stage B Prospective Phase Perturbation

Stage B is a future causal protocol requiring separate ethics, safety, equipment, and acquisition approval. It should:

1. localize a receiver-specific beta phase-transfer curve independently;
2. randomize trials among open-phase, closed-phase, amplitude-only, and sham conditions;
3. match carrier frequency, pulse count, charge, energy, and cue timing;
4. use receiver spike/evoked-response transfer in a fixed 10-80 ms window as the primary endpoint;
5. treat task error and reaction time as secondary endpoints; and
6. test rescue by shifting receiver integration state and predicting a reversal of the preferred phase.

The primary causal estimand would be

$$\text{ATE}_{\text{phi}} = E[Y \mid \text{do}(\text{delta}_{\text{open}})] - E[Y \mid \text{do}(\text{delta}_{\text{closed}})].$$

The provisional planning floor is 100 analyzable trials per condition, retained only if a pre-acquisition clustered simulation verifies at least 80% power for the chosen effect. Computational phase scrambling and feature deletion in Stage A are ablations, not biological interventions.

11 Falsification Policy

The Stage A claim fails for this preparation if any of the following holds:

1. monkey N places Δ_{TNR} inside the ± 0.02 -bit equivalence interval;
2. M5-linear-LFP or M5b-nonlinear-LFP matches or exceeds M6-TNR under the locked compute rule;
3. the increment vanishes after evoked-waveform, power, common-input, or spatial-embedding controls;
4. the effect changes sign across the bounded reference/phase-estimator sensitivity set;
5. phase variables fail the preregistered reliability gate;
6. only a post hoc band, time window, feature subset, or animal works; or
7. the locked pipeline cannot be reproduced from the frozen data and configuration.

A Stage B null selective phase effect would reject the proposed phase-specific causal transfer in that preparation. Neither Stage A nor Stage B alone adjudicates all oscillatory routing, all neural reconstruction, or consciousness.

12 Relation To SAN, NAPOT, And Micah's Genealogy

TNR is an empirical bridge extracted from the longer Self-Aware Networks (SAN) and Neural Array Projection Oscillation Tomography (NAPOT) genealogy. The documented sequence is not compressed into one retrospective date:

- **2011-2012:** predictive self/world modeling, recurrent feedback, neural computation, and slower/faster oscillatory structure appear in the early SAN dialogues;
- **2017:** the Neural Lace and 3-D-shape discussions make the spatial-predictive and multisensory read/write bridge explicit;

- **2022:** the GitHub corpus names NAPOT and develops distributed projection, waveform, phase, reader-relative, and tomography claims;
- **2024 onward:** later books and papers organize those atoms into explicit comparator and falsification programs.

The paper-relevant source ladder is more precise than those four summary bullets:

Stage	Date and date type	Bounded TCNR-relevant contribution	Provenance boundary
Early stereographic-camera note	May 13, 2011 owner/internal date; first immutable Git fixation June 12, 2022	Distributed “pixels” contribute partial scene information; statistical relations support inference and prediction; senses combine into a continuously updated multisensory picture.	The 2011 date is preserved as an internal authorial date, not represented as a 2011 public Git date. The immutable public artifact is the 2022 commit [24].
Public projector dialogue	August 23–24, 2012 service/public timestamp	The mind is described as a projector that predicts the world and supplies much of the experienced detail internally.	Functional ancestor of predictive neural rendering; it does not contain the mature TCNR operator [25].
Oscillatory conscious-canvas post	July 24, 2013 platform-displayed date in a hash-frozen 2026 Facebook screenshot; direct public-audience object route	Sensory experience is described as cycling synchrony; learned oscillation patterns can be reactivated to render vectors, lines, sounds, and a world; smaller oscillations are described as combining into larger brainwaves that “knit the canvas of conscious experience.”	This is an early oscillatory-rendering and multiscale-composition ancestor. It does not backdate PWD, NAPOT, TCNR, projection-slice mathematics, or the SAN Tomogram. Facebook edit history and original response-body custody remain open [30].
Neural Lace Podcast	April 11, 2017 public timestamp	Spatial computation, a time-varying three-dimensional ionic matrix, feedback, prediction, and a brain-network protocol question are joined in one public discussion.	Public functional ancestor; it does not backdate NAPOT, PWD, or TCNR terminology [26].
NAPOT Git and public explainers	June–September 2022 immutable commits and service/public timestamps	Receive-transform-project arrays, distributed observation, phase-sensitive rendering, projection-slice and volumetric reconstruction analogies, memory scaling, and sensorimotor updating become an explicit joined program.	This is the mature public SAN/NAPOT source layer; biological validation remains separate [27,28].
Book consolidation	October 2024 published book and public Git records	Molecular, synaptic, dendritic, oscillatory, rendering, and action levels are consolidated into one sustained SAN account.	Published synthesis, not an empirical validation of TCNR [29].
TCNR4	October 19, 2025 embedded artifact date; November 15, 2025 immutable public Git fixation	The render operator, TCI proposal, six-condition gate, cross-substrate tests, SIT bridge, Telos, and optional Limit Mind are stated as one TCNR program.	Artifact date and public-fixation date are distinct. The exact 39-page PDF is hash-bound [23].
The mathematical analysis successor	July 27, 2026 current synthesis	The empirical protocol, historical conceptual program, corrections, failure conditions, and provenance ladder are preserved in one lossless but evidentially separated manuscript.	Current synthesis; it does not move any earlier date or report an empirical result.

The historical priority record and the scientific test have different jobs. The genealogy establishes what Micah publicly described and when. Stage A tests one present-day operational residual. A negative result would not erase the dated record; a dated record would not convert a negative experiment into a positive one.

The Facebook recovery also creates a later-work comparison opportunity, including comparison with public QRI wave, field, attractor, replay, and Fourier-family documents. That comparison remains a separate,

proposal-only audit. Its row counts and author-navigation edges are not scientific evidence, do not establish access or dependence, and do not alter this paper’s constructive argument. Public QRI sources may enter later drafts only as passage-bound scientific or mathematical comparators.

An earlier full manuscript in this line, *TCNR4*, is the direct documentary predecessor to the present protocol [23]. Its embedded PDF metadata records creation on October 19, 2025. The exact 39-page artifact was fixed publicly in Git on November 15, 2025, in commit `eab1ea65c3abd488c2a33e6bf4f420c32951fcc3`. These facts establish artifact provenance and public availability. They do not independently validate TCNR’s biological, computational, consciousness-related, cross-substrate, or metaphysical hypotheses.

13 Part II. Firewalled Conceptual Bridge

Part II — Difference, Pattern, Action, Consciousness, And Cross-Scale Boundaries

14 From Difference To Pattern

The starting intuition is that a pattern is not one isolated element. It is a structured set of differences and relations among elements. Let the elements at time t be vertices $V_t = \{x_1, \dots, x_n\}$. For a declared measurement domain, define a relational difference

$$\Delta_{ij,t} = d(x_{i,t}, x_{j,t}), \quad (1)$$

where d may be Euclidean, circular, graph-geodesic, probabilistic, semantic, or another predeclared metric. A pattern candidate is then

$$P_t = (V_t, E_t, \Delta_t, C_t), \quad (2)$$

where E_t specifies which relations are eligible and C_t contains a bounded coherence or compatibility statistic defined for that domain. This notation does not say that every difference is neural phase or quantum coherence. It says only that isolated values become a relational object when their comparison rule, eligible edges, and measurement scale are declared.

Information theory already warns against equating a vivid pattern with information in the abstract. Information is defined relative to a distribution and an uncertainty reduction, not by aesthetic regularity alone [7]. Two coincident marks can be informative if they change the receiver's probability model, but coincidence does not automatically equal one bit. Likewise, a relation can be useful without being oscillatory, and oscillatory coherence can be high while carrying no task-specific information. TCNR therefore requires separate held-out estimates for discriminability, prediction, coherence, and action consequence.

The reconstruction map is

$$R(P_t) \rightarrow \hat{s}_t, \quad (3)$$

where $\hat{s}_t$ is a latent state or calibrated task-relative estimate. A part need not resemble the recovered whole. Many different projections can jointly constrain a latent object even when no single measurement contains the object in finished form. This is the defensible content of the jigsaw and tomography analogies. Literal tomography requires an identified forward operator, nonredundant projections, observability or identifiability conditions, a reconstruction criterion, and an operator-specific error test. Without those conditions, *relational recurrent reconstruction* is the narrower term.

This requirement is consistent with multiview nonlinear-ICA results: several sufficiently distinct views can make a common latent source identifiable under stated independence, corruption, and invertibility assumptions, while a single arbitrary nonlinear mixture is not uniquely invertible [60,61]. The Qualia Tomogram application must test its actual assumptions rather than cite those theorems as a blanket guarantee.

14.1 Receiver-relative observability and mixture identifiability

A physical difference need not be created by an observer in order to matter. The receiver-relative question is narrower: can the difference affect the observations or actions available to a declared receiver? Let a physical state X generate a measurement Y through

$$p(y \mid x), \quad \text{or} \quad y = H(x) + \epsilon. \quad (4)$$

Two physical states may differ while producing indistinguishable measurement distributions for one channel. Conversely, a channel may separate them reliably. TCNR therefore uses *receiver-relative* to mean accessible, discriminable, or causally usable through a specified observation-and-action interface. It does not mean that observation creates the underlying physical difference.

This boundary is also an inverse-problem constraint. If one observed composite satisfies

$$y = F(x_1, x_2), \quad (5)$$

and several pairs (x_1, x_2) generate the same y , the latent components are not identifiable from that observation alone. Recovering the components requires additional channels, interventions, independence or sparsity assumptions, a calibrated forward operator, or other constraints that can themselves be tested. A compelling composite percept is therefore evidence about the receiver's render, not automatic evidence that the receiver has recovered the composite's unique causal history. This is why TCNR evaluates task-relative prediction and perturbation while refusing to call an unconstrained latent decomposition literal tomography.

14.2 Four meanings of coherence

The word *coherence* is overloaded. TCNR must keep at least four meanings separate.

Domain	Operational meaning	Suitable observable	What cannot be inferred
Statistical dependence	Variables depart from a declared independence model.	Mutual information, likelihood ratio, covariance, conditional dependence.	Phase locking, causal direction, or consciousness.
Neural oscillatory coherence	Band-limited signals have a reproducible cross-spectral phase/amplitude relation at a declared scale.	Complex coherence, imaginary coherence, phase-locking value, directed model with controls.	Quantum coherence, semantic agreement, or sufficient integration.
Computational integration	Several inputs jointly improve a latent estimate or policy beyond their parts.	Out-of-sample synergy, ablation, reconstruction error, control gain.	A particular oscillatory implementation or subjective experience.
Quantum coherence	Off-diagonal structure relative to a specified basis survives in a density operator and supports interference under controlled dynamics.	Quantum-state or process tomography and resource-theoretic measures [11,12].	Macroscopic mental unity, neural phase synchrony, or scale invariance.

The same mathematical word can appear in these rows without making the physical mechanisms identical. A cross-scale theory needs an explicit translation map, dimensional consistency, and a test that could fail at each bridge.

A single temporal coincidence is not yet a coherence estimate. Statistical or oscillatory coherence requires repeated observations, a declared estimator, and comparison with a null that preserves the relevant marginal

structure. Stable classical phase locking can arise from direct coupling, common input, filtering, reference choice, or shared task drive; it does not by itself establish causal direction or additional task information.

14.3 Surprise, clarity, meaning, and causal value

A rare event is not automatically meaningful, true, useful, or clear. **Conditional surprise**, or conditional self-information, can be written

$$I_t = -\log_2 p(y_t \mid y_{<t}, a_{<t}), \quad (6)$$

but high surprise can arise from noise, sensor failure, or an irrelevant outlier. Bayesian information gain instead measures the change between prior and posterior beliefs,

$$IG_t = D_{KL}[q(s_t \mid y_{\leq t}) \parallel q(s_t \mid y_{<t})], \quad (7)$$

and still does not guarantee semantic relevance or action value [17]. TCNR therefore separates:

- **surprise**: improbability under a model;
- **discriminability**: separability at a receiver;
- **information gain**: posterior change;
- **clarity**: estimator reliability or uncertainty;
- **semantic relevance**: relation to a task or representational variable; and
- **causal value**: change in prediction or action under intervention.

A phasic event counts as evidence for TCNR only when it improves a held-out endpoint or changes a declared receiver/action variable under the relevant controls. A bright spectral event or rare coincidence is not sufficient.

14.4 Pattern tokens, pattern types, and persistence

Claims that an object or technology is an evolving pattern can refer to several different things:

1. one physical token persists while its microstate changes;
2. a type recurs in many different physical tokens;
3. a cultural or technological lineage transmits constraints across tokens;
4. a system causally repairs or reproduces itself; or
5. an observer groups different tokens under one concept.

A lamp design recurring across centuries is mainly type-level and cultural continuity. It is not one physical state literally persisting, and it does not show intrinsic agency. This distinction is important for cross-substrate TCNR: structural recurrence, self-maintenance, reproduction, prediction, and counterfactual action are separate criteria.

14.5 Coherence-matrix determinant correction

The original TCNR source treated positive $\log \det(C + \epsilon I)$ as a generic coherence reward. That direction is not valid without a specific interpretation. For a two-variable normalized correlation matrix

$$C = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \det C = 1 - \rho^2, \quad (8)$$

the determinant decreases as $|\rho|$ increases. A positive log-determinant can therefore reward independence or spectral volume rather than synchrony. The present protocol does not use that term as its primary endpoint. Any future multivariate summary must declare whether it measures dependence, diversity, effective dimensionality, or precision; establish its monotonic direction on simulated cases; and compare it with direct coherence/phase-locking and predictive endpoints. No scalar determinant is a consciousness measure.

15 From Pattern To Prediction And Action

Recognition alone does not close the TCNR loop. A reconstructed state matters computationally when it changes prediction, policy, or control. Let b_t be the agent or controller state, a_t an action, and y_{t+1} the next observation. A minimal loop is

$$\hat{s}_t = R(P_t), \quad a_t \sim \pi(a \mid \hat{s}_t, b_t), \quad P_{t+1} = U(P_t, a_t, y_{t+1}). \quad (9)$$

This yields the causal sequence

$$\begin{aligned} &\text{differences} \rightarrow \text{relational pattern} \rightarrow \text{latent estimate} \rightarrow \text{predicted consequence} \\ &\quad \rightarrow \text{selected action} \rightarrow \text{feedback} \rightarrow \text{updated pattern.} \end{aligned} \quad (10)$$

Recurrent contribution is not inferred merely because activity persists. Trial-residual dynamics can be used to separate recurrently generated contributions from condition-averaged drive under an explicit model [62]. Movement and arousal are equally important alternatives: spontaneous behavior and richly varied movements can dominate widespread single-trial neural activity [54,55]. Every application in this paper therefore carries measured movement/state covariates and a movement-matched negative control.

Active-inference and predictive-processing frameworks provide important prior architectures for linking generative models, perception, planning, proprioception, and action [8-10]. Their broad explanatory reach is not itself proof of TCNR. Recent reviews emphasize that empirical adequacy requires comparisons against alternative models and qualitatively distinct predictions [9,10]. The TCNR residual is narrower: after the same raw data, history, capacity, and compute are supplied to the strongest comparator, do declared relational phase-geometry variables improve prediction or control, and does selective phase perturbation change receiver transfer?

Intention should not be reduced to reflex. A state-dependent response may be intentional only if the system represents or realizes alternatives in a way that affects policy. TCNR's counterfactual capacity condition is meant to prevent a fixed oscillator or stimulus-response chain from being called deliberative merely because it is coherent. Candidate operationalizations include policy diversity, local controllability, or behavior under counterfactual cue changes. None is yet a validated consciousness criterion.

16 Consciousness Science: What TCNR Can And Cannot Test

Contemporary consciousness theories address different targets: phenomenal character, conscious access, report, metacognition, global availability, recurrent processing, higher-order representation, predictive self-modeling, and causal integration. They are not interchangeable [13-15]. TCNR should therefore avoid a single scalar *consciousness index* until a target construct, measurement model, and external validation are established.

The original TCNR source proposed a weighted TCI combining predictive mutual information, coherence, counterfactual capacity, and prediction error. That formula is retained as historical design material, not as a validated measure. A weighted sum can hide compensation: high coherence might numerically offset absent counterfactual behavior even if both were claimed necessary. It can also mix estimators with different reliability, units, and causal status. The present paper replaces that scalar with a **claim vector**:

$$X_t = (I_{pred}, E_{recon}, C_{rel}, K_{cf}, G_{access}, S_{self}), \quad (11)$$

where:

- I_{pred} is held-out predictive information or a proper-score improvement;
- E_{recon} is operator-specific reconstruction error;
- C_{rel} is a declared relational/coherence statistic;
- K_{cf} is counterfactual policy or controllability capacity;
- G_{access} is a separately measured access, report, or task-availability endpoint; and
- S_{self} is an independently defined self/body/agency endpoint when relevant.

No component is silently imputed, and no weighted total is called consciousness. Different studies may test different arrows. Stage A in this manuscript tests only a representation-utility residual for a cue-instructed grip condition. It does not measure G_{access} or S_{self} and is not a consciousness experiment.

A future consciousness-relevant TCNR program would need at least:

1. a declared construct such as access, metacognition, or state transition;
2. matched attention, memory, decision, report, and motor controls;
3. a phase-structure model compared with equal-input alternatives;
4. a causal intervention with a construct-specific endpoint;
5. negative conditions in which coherence exists without the target construct; and
6. replication across tasks or measurement contexts.

Even success would support a mechanism contributing to the declared construct, not a proof that TCNR is necessary or sufficient for phenomenal experience.

16.1 Intersubjective render convergence

Different people can report similar experiences or memories because they share sensory evidence, language, cultural priors, leading questions, source-monitoring errors, or post-event discussion. Classic false-memory and memory-conformity experiments show that structured convergence can be produced without a shared percept or altered history [18,19]. TCNR can turn this into a prospective social reconstruction test:

1. elicit independent reports before discussion;
2. manipulate whether participants receive common versus distinct cues;

3. vary post-event social exposure;
4. model semantic attraction and source confidence;
5. compare individual and group-level latent reconstructions; and
6. test whether convergence tracks shared evidence and transmission rather than an unexplained common internal state.

The permitted term is **intersubjective report convergence** or **collective false memory** when the content is false. Similar reports do not establish a literally shared hallucination, alternate timelines, collapsed universes, or retroactive changes to the past.

16.2 Conscious time and computational event rate

The intuition that more computation per physical second could change experienced time is a useful question, but it divides into several endpoints:

- physical elapsed time;
- neural or computational event rate;
- online duration judgment;
- temporal resolution or discrimination;
- retrospective remembered event density; and
- phenomenal duration.

These can dissociate [20]. An external processor could increase prediction depth or memory density without proportionally lengthening online experienced duration. A valid experiment would manipulate processing or event density while separately measuring prospective duration judgment, retrospective duration, temporal discrimination, memory count, performance, arousal, and report strategy. TCNR makes no present claim that computation rate is consciousness rate.

16.3 Historical TCI and six-condition program

The source record must not disappear merely because its evidential ceiling changed. TCNR4 proposed a Tomographic Consciousness Index (TCI) that combines four null-standardized components: predictive information, a band-limited coherence summary, counterfactual capacity, and negated prediction error. It also proposed a six-part gate covering band-limited coherence, lawful traveling-wave routing, closed feedback, memory/hysteresis, counterfactual agency, and dynamic inference [23]. In TCNR4 these conditions were presented as jointly necessary rather than sufficient; a later metaphysical companion used stronger language. The mathematical analysis preserves the original proposal while separating three statuses:

1. **historical specification:** the formula, component list, nulls, and gate belong to the dated

TCNR4 record;

1. **current measurement program:** each component can be estimated and tested separately through the claim vector in Section 15; and

1. **unvalidated interpretation:** neither $TCI > 0$ nor passage of the six-part gate is presently established as a criterion for consciousness.

The log-determinant correction in Section 13.5 is therefore an additive repair, not deletion of the coherence

term. A future TCI successor may use a directionally validated dependence, diversity, or effective-dimensionality statistic, but it must calibrate the construct against positive and negative consciousness-relevant conditions and must prevent one component from numerically hiding failure of another. The historical TCI, its benchmark idea, and its adversarial nulls remain available as a research program.

17 Physics Bridge: Local, Conditional, And Testable

The physics language in TCNR is motivated by a legitimate observation: neural signals are physical, phase and frequency are physical variables, and reconstruction ultimately occurs in a material substrate. The bridge becomes unsafe when similar words are treated as identities. Neural oscillatory coherence is not quantum coherence. Classical simultaneity is frame-dependent rather than an observer-free statement that two distant events happen at the same time. A local field interaction cannot be inferred from matching patterns at arbitrary separation.

17.1 Scale invariance is not scale identity

A relation may have the same dimensionless form at several scales while its carrier, noise, coupling, energy, and causal range differ. If a model claims scale invariance, it should specify a transformation $\mathcal{S}_\lambda$ and the quantities preserved:

$$\mathcal{M}[\mathcal{S}_\lambda(x)] = \mathcal{T}_\lambda(\mathcal{M}[x]). \quad (12)$$

The claim is empty until $\mathcal{S}_\lambda$, $\mathcal{T}_\lambda$, and the error tolerance are defined. Locality is a separate condition: interaction strength, delay, or influence must be a function of physical or graph distance under a declared propagation law. Two far-separated events can be statistically coincident without exchanging a signal or forming a physical unit.

17.2 Quantum coherence and decoherence

In quantum theory, coherence is basis-relative and described through a density operator; resource- theoretic measures and allowable operations make the definition explicit [11,12]. Decoherence is an open-system process in which environmental interaction suppresses interference in a selected basis; it is not a synonym for disagreement, randomness, neural desynchronization, or a pattern boundary. The TCNR difference graph may use *coherence* and *decoherence* as bounded relational labels only if the measurement domain is named. Quantum terminology requires an actual quantum state/process model and observables that distinguish it from a classical stochastic account.

Classical synchronization is also not quantum entanglement. Matching frequencies, a stable relative phase, or correlated responses can be represented by classical coupled oscillators. Quantum entanglement concerns nonseparability of a joint quantum state and requires an actual quantum-system description plus a valid entanglement witness, state/process reconstruction, or another certified criterion [22]. Bell-inequality violation is one stronger device-independent route in suitable experiments, not a prerequisite for every entanglement claim. Persistence of classical phase locking until perturbation does not make the signals entangled, and loss of neural synchrony is not quantum disentanglement.

17.3 Spacetime, computation, and information

The philosophical proposition that spacetime is computational can be separated into at least three claims:

1. physical evolution can be represented computationally;
2. the universe literally implements a computation; and
3. information or computation is ontologically fundamental.

The first is methodologically ordinary. The second requires a mapping from physical states to computational states and a nontrivial account of implementation. The third is metaphysical. TCNR does not establish claims 2 or 3. A neural reconstruction result supplies no evidence that spacetime itself is a mind or computer.

The more defensible bridge is structural: physics, neuroscience, and artificial systems can each be described using states, relations, transformations, observables, and interventions. Structural similarity can motivate a formal translation, but only empirical adequacy and bridge-specific tests justify identification. This stance is compatible with structural-realist questions while remaining neutral about whether relations or objects are ontologically fundamental [16].

17.4 Time asymmetry and thermodynamic language

The original source also linked coherence flux, memory, and a time-density motif to the arrow of time. A defensible stochastic-thermodynamic bridge begins with forward and reversed path probabilities. Under suitable conditions,

$$\Sigma = D_{KL}(P_F||P_R) \geq 0 \quad (13)$$

quantifies path-space irreversibility or entropy production [21]. The time derivative of an arbitrary KL divergence is not automatically nonnegative and is not automatically a physical entropy-production rate. Boundary conditions, coarse graining, reservoirs, units, and the reverse process must be defined.

In TCNR, **time density** may be used operationally for an event, update, or memory-encoding rate per declared clock interval. It is not identified with relativistic proper time or a gravitational source unless a separate dimensionally consistent theory supplies and tests that bridge. Neural hysteresis can support a memory trace without proving a cosmological arrow of time.

17.5 Historical SIT, cosmic, teleonomic, and metaphysical bridges

TCNR4 also placed the local render loop inside a larger Super Information Theory (SIT) vocabulary: coherence fields, an informational-event-rate or “time-density” variable, informational energy, memory as hysteresis, and a forward arrow associated with dissipative history [23]. It then introduced a teleonomic objective—called Telos—in which a renderer’s policy trades integrated prediction and valued outcomes against unnecessary dissipation. Finally, it described an optional cosmic limit construction: maximize a corrected TCNR objective over physically admissible render-stitch systems, examine limiting sequences, and ask whether a maximally capable renderer could support a separate Principle-of-Sufficient-Reason or modal argument.

Those propositions are preserved here because they are part of Micah Blumberg’s authored argument, but they do not inherit evidence from Stage A. Four different research questions must be kept distinct:

1. **computational teleonomy:** does a declared objective explain or predict adaptive policy under resource constraints?

1. **thermodynamic realization:** can objective terms be mapped to measured work, entropy production, or dissipation with consistent units and boundary conditions?

1. **cosmic generalization:** is there an identified physical state, observation map, renderer, and intervention at the proposed scale?

1. **metaphysical inference:** do independent premises such as sufficient reason or modal logic support a conclusion after the empirical claims are removed?

The local neuroscience protocol answers none of questions 2–4 and only motivates a bounded version of question 1. The SIT, Telos, cosmic-renderer, and metaphysical constructions therefore remain historical TCNR4 propositions and companion-paper routes. They are not retracted, silently erased, or represented as results of the macaque analysis.

18 Cross-Scale And Cross-Substrate Program

TCNR is not licensed to move from cortex to plants, biofilms, machines, or cosmology by analogy alone. A cross-substrate claim requires the same typed causal interface:

$$\begin{aligned} \text{partial observations} &\rightarrow \text{declared relational transform} \rightarrow \text{latent estimate} \\ &\rightarrow \text{policy/action} \rightarrow \text{feedback and adaptation.} \end{aligned} \tag{14}$$

For each substrate, the study must name:

- the state variables and observation operator;
- the receiver or integrator and its physical scale;
- the relational/coherence estimator and nulls;
- the action or control endpoint;
- the counterfactual intervention;
- the adaptation timescale; and
- an equal-input alternative model.

The cross-scale hypothesis is supported only when the typed operation and discriminating residual survive in more than one substrate. It is rejected or narrowed when the apparent match depends on metaphor, inconsistent units, outcome-selected bands, or a missing action/feedback arrow. A universe-level application is a philosophical limit question until those fields and interventions are defined; it is not an empirical conclusion of the macaque protocol.

TCNR4 specifically proposed plant, biofilm, artificial-oscillator, clinical, and perturbational benchmark routes [23]. The mathematical analysis preserves those routes as prospective hypotheses:

- a substrate should expose partial observations rather than only a finished global state;
- a measurable relational transform should improve prediction or control beyond equal-input

alternatives;

- phase, amplitude, power, common drive, geometry, and generic model capacity should be separated;

- perturbing the proposed relation at matched energy should alter a receiver or behavioral endpoint; and
- high synchrony without repertoire, adaptation, or task benefit should count as an adversarial false positive.

No default extension is granted. Each substrate must earn the TCNR label by passing its own measurement, comparator, perturbation, and failure conditions. This converts the original cross-substrate assertion into a preserved and falsifiable research program rather than deleting it.

19 Non-Claims And Limitations

This protocol does not claim that:

- cue-instructed grip-condition decoding is consciousness or a cue-independent maintained plan;
- coherence or beta phase is universally necessary;
- a positive decoder proves a volumetric neural image;
- a planar gradient uniquely identifies a traveling wave;
- phase differences assign causal edge direction;
- motor-cortex findings generalize to vision without a separate dataset;
- plants, biofilms, or artificial systems instantiate the same mechanism;
- TNR validates all of SAN, NAPOT, SIT, or QGTCD; or
- a null result in two animals disproves phase-dependent computation everywhere;
- neural phase coherence is quantum coherence;
- classical coincidence, matched frequency, or phase locking is quantum entanglement;
- receiver-relative observability means that an observer creates the underlying physical difference;
- one composite measurement uniquely identifies its latent causes without a forward model or added constraints;
- statistical coincidence implies physical interaction or causal influence;
- scale-invariant notation makes mechanisms identical across scales;
- spacetime is empirically shown to be a computer, mind, or consciousness substrate; or
- the conceptual difference-pattern-action ladder validates a consciousness index.

The two-animal sample limits population generalization, and their different M1/PMd versus lateral/PMv coverage prevents a direct anatomical-replication claim. The same animals cannot provide both model discovery and a large independent population estimate. The fixed public data may be underpowered at the declared smallest effect. Task cues, evoked responses, movement preparation, volume conduction, reference choice, cycle shape, and array geometry are serious alternatives addressed by controls but not metaphysically eliminated.

20 Data, Code, Ethics, And Release Status

The source experiment's ethics and animal-care approvals are reported by Brochier et al. This protocol performs secondary analysis of public data and introduces no new animal procedure. Stage B would require new approval.

No large data file is redistributed here. The exact repository commit and annex identities are recorded so an

authorized user can retrieve the source from G-Node under CC BY 4.0. Analysis code, environment lock, downloaded-file hashes, and results will be released only after the pipeline is implemented, replayed, and reviewed.

This The mathematical analysis is a locally final-preprint-ready, adversarially integrated theory-and-protocol successor. Part I preserves the prospective neural test; Part II is a conceptual bridge with no empirical authority over that test. The paper contains no empirical result. The exact local package is suitable for an author-controlled preprint-version decision, but this manuscript does not itself perform a Zenodo deposit, OSF deposit, DOI action, upload, or other provider mutation.

Part III — The Qualia Tomogram Generalization And Proof Program

21 The Problem Before The Qualia Tomogram

Consider a familiar object such as broccoli on a plate. Vision supplies color, contour, depth, and location. Touch supplies texture, resistance, and temperature. Smell and taste supply different molecular relations. Proprioception and the vestibular system locate the body that approaches the plate. Memory supplies learned expectations; action tests them. No receptor receives broccoli as a finished token, and no one neural measurement contains every relation in the completed experience. Color can be decoded and reconstructed from distributed human visual-cortex responses, but that result does not imply that one recording site contains the completed experience or that a decoder is already a causal write interface [35].

Merely collecting the partial signals would produce a list. A usable percept requires the nervous system to determine which relations belong together, which remain separate, where they lie relative to the body, which remembered relations apply now, and what action should follow. A candidate solution must also predict what the participating receivers will register next. If an alleged reconstruction fits only the signals used to create it, it may be a flexible description rather than the operation used by the organism.

This yields the governing problem:

Can several conditionally distinct biological views constrain one changing body-world state, and can

the declared relation be distinguished from rate, power, common drive, movement, report, model

capacity, and an unconstrained nonlinear decoder?

The solution proposed here is recurrent projection consistency. Each receiver preserves and transforms part of the event according to its learned structure and current state. A candidate body-world estimate is projected through those declared receiver models. Mismatch changes the estimate, receiver state, route, or action. Repetition tests the reconstruction against new sensory evidence. Only after this operation is understood do the names become useful: a receiver-relative consequential departure is a Phase-Wave Differential (PWD); **the receive-transform-project-recur architecture is Neural Array Projection Oscillation Tomography (NAPOT); and the active distributed reconstruction is a SAN Tomogram.**

The SAN claim is stronger than decoding. The tomogram is proposed to be the physically active form of the qualitative scene for the participating network, not a code displayed to a hidden observer. That constitutive claim receives no authority from reconstruction accuracy alone. It must survive content-specific dissociations, perturbations, rescues, and no-report or delayed-report controls.

22 Classical Projection-Slice Mathematics And The Biological Generalization

Classical tomography earns reconstruction through a known forward model. For a two-dimensional object (f), a common form of the projection-slice theorem is

$$\mathcal{F}_1\{\mathcal{R}_\theta f\}(\omega) = \mathcal{F}_2\{f\}(\omega \cos \theta, \omega \sin \theta). \quad (15)$$

The equation relates the one-dimensional Fourier transform of a Radon projection to a slice through the two-dimensional Fourier transform of the object. It demonstrates that partial measurements can preserve lawful information about a larger state. It does not make neural projections line integrals, fix their geometry, or identify the inverse when the biological operators are unknown [31,32].

Fourier decomposition answers which frequency components compose a signal. The fractional Fourier transform parameterizes intermediate rotations through a time-frequency or space-frequency plane under a declared convention. Projection-slice reconstruction asks how distinct projections constrain an object. Recurrent state estimation asks how a changing latent state is inferred through time. These operators can be compared or composed, but vocabulary does not make them mathematically identical [33,34]. Point-set classification and neural view synthesis provide useful engineered comparators for partial-observation integration, yet neither PointNet nor NeRF is treated as a biological mechanism or a consciousness model [36,37].

The biological problem is harder than the classical special case. Receptors, dendrites, cells, arrays, and body states are learned and history-dependent. Their observation maps may be nonlinear, recurrent, time-varying, only partly measured, and non-identifiable. Accordingly, receiver (i) at time (t) is modeled as

$$y_{i,t} = P_{i,t}(q_t; h_{i,t}, m_t) + \varepsilon_{i,t}, \quad (16)$$

where (q_t) is a candidate latent body-world state, $(h_{i,t})$ is learned and recent receiver history, (m_t) is bodily and modulatory context, and $(\varepsilon_{i,t})$ collects declared measurement and model error. Every application must say what $(P_{i,t})$ means biologically rather than allowing it to absorb unexplained flexibility.

23 Receiver-Relative Difference And Generalized Reconstruction

A receiver is not blank when a new event arrives. Membrane state, dendritic conductance, inhibition, recent firing, synaptic history, neuromodulation, and network context affect the event's consequence. Let

$$d_{i,t} = T_i(y_{i,t}, \mathbb{E}[Y_i | c_{i,t}]) \quad (17)$$

denote a typed difference between the current observation and the receiver-conditioned expectation $(c_{i,t})$. The transform (T_i) may expose declared timing, circular phase, duration, waveform, magnitude, route, or release variables. A coefficient of variation or phase-dispersion statistic can be one component of $(d_{i,t})$; neither is synonymous with PWD, Shannon information, or biological meaning.

For candidate reconstruction (q) , define

$$\hat{q}_t = \arg \min_q \left[\sum_i w_{i,t} \ell_i(P_{i,t}(q), d_{i,t}) + \lambda \Omega(q) \right], \quad (18)$$

where the loss (ℓ_i) , reliability weights $(w_{i,t})$, and regularizer or biological prior (Ω) are declared before the held-out test. A squared error is one special case, not a universal biological law. The equivalence class of observations that the declared receiver family cannot separate is

$$[q]_{\mathcal{P}} = \{q' : P_{i,t}(q') = P_{i,t}(q) \text{ for every observed } (i, t)\}. \quad (19)$$

If this class contains several states, the current data do not identify a unique reconstruction. More receivers, interventions, structural assumptions, or uncertainty reporting are required. This non-identifiability result is not a weakness to hide; it is the exact boundary between a disciplined tomographic claim and an unconstrained latent metaphor.

Projection consistency is evaluated on observations withheld from fitting:

$$E_{\text{hold}}(\hat{q}) = \sum_{(i,t) \in \mathcal{H}} w_{i,t} \ell_i(P_{i,t}(\hat{q}), d_{i,t}), \quad (20)$$

with separate reports for held-out receiver, condition, modality, session, day, subject, perturbation, and rescue. Training error is never substituted for these endpoints.

24 The Working Application Proof

The paper will include a reproducible generalized-tomography reference application rather than stop at an architectural diagram. Its first vertical slice will generate or load a known latent multimodal state, pass it through declared state-dependent partial-observation operators, reconstruct from a strict subset, and predict the omitted observations. All data splits, seeds, operator classes, optimizer budgets, and failure cases will be frozen before confirmatory scoring.

The minimum matched model set is:

1. mean or prevalence baseline;
2. local amplitude/rate/power summary;
3. scalar coherence without spatial or receiver structure;
4. linear state-space model;
5. classical known-geometry tomography where applicable;
6. generic recurrent model with the same observations;
7. flexible nonlinear model with matched capacity and compute; and
8. the declared receiver-relative generalized-reconstruction model.

For model (M) and the strongest locked comparator (B), the primary predictive increment is

$$\Delta_{M:B} = L_{\mathcal{H}}(B) - L_{\mathcal{H}}(M), \quad (21)$$

using a proper held-out loss and a prespecified smallest effect of interest. Parameter count, effective capacity, input access, tuning opportunities, and compute are reported rather than hidden behind one label.

The broccoli laboratory supplies the multimodal application. Separate channels represent color, form, texture, odor/taste proxy, body-relative position, expectation, and action affordance. The benchmark asks whether the model can reconstruct the shared object while preserving the channel distinctions; predict a withheld relation; adapt when the carrier implementation drifts; and detect an artificial insertion without turning the inserted feature into the whole self-model. This is a bounded computational proof of concept, not evidence that a human participant was made to experience broccoli.

The inherited TCNR motor-cortex protocol remains a second application branch. Its Brochier/G-Node data, development-versus-replication animal split, cue/grip confound, causal phase estimator, comparator ladder, surrogates, smallest effect of interest, and no-result boundary are preserved until the analysis is actually implemented or a documented feasibility audit establishes why it cannot be completed.

25 Perturbation, Rescue, Drift, And Action Return

Feature deletion tests whether an analysis uses a variable; it is not a biological intervention. The application therefore labels computational ablation, simulated causal intervention, and biological perturbation separately. For declared projection (j), define

$$\Delta E_j = E_{\text{hold}}^{\text{intervene}(j)} - E_{\text{hold}}^{\text{matched control}}. \quad (22)$$

A valid selective prediction states which latent relation, perceptual endpoint, receiver response, or action should change. A matched rescue restores the declared relation while matching nonspecific drive, energy, timing, arousal, and motor readiness. Rescue is specific when the predicted component recovers more than off-target components and more than sham or generic activation.

Carrier drift provides a deeper test. If the content survives while the participating cells or channels change, the model should preserve the relevant relation without relying on fixed unit identity. If a fixed-cell classifier fails but a relational model succeeds on held-out drift, that supports carrier-flexible reconstruction within the declared simulation or dataset. It does not establish phenomenal persistence.

Neural drift is not one universal law: longitudinal studies report dynamic reorganization, stable task information despite unstable cells, stimulus-dependent visual drift, and cases where latent population dynamics remain stable over long intervals [48–52]. The benchmark must therefore measure both unit-level change and relation-level stability instead of assuming in advance that every carrier drifts.

The loop closes in action:

```
partial observations
→ receiver-relative transformations
→ recurrent reconstruction
→ object and body-relative separation
→ predicted consequence
→ selected internal or motor action
→ returned sensory evidence
→ reconstruction update and plasticity
```

A reconstruction that never changes prediction, choice, action, or later reception remains descriptively interesting but computationally underdetermined.

26 Memory Reconstruction And The Conscious Boundary

External perception and memory reinstatement can be tested under the same forward-model discipline. A partial cue activates a present carrier set; the candidate reconstruction predicts withheld features, receiver relations, and consequences. Same-content realizations across carrier drift should remain closer under the

declared relational geometry than changed-content controls. The *Virtual Engrams* paper retains the complete memory-stability program; this manuscript uses one reconstruction and loss/rescue test.

Optogenetic engram reactivation and false-memory experiments establish that experimentally selected hippocampal ensembles can drive later fear-related behavior under specific preparations [58,59]. They do not show that the manipulated ensemble is the complete remembered experience or that behavioral recall identifies the original qualitative state.

Availability in memory is not current conscious presence. A reconstructed candidate enters the active SAN Tomogram only if it develops the declared reciprocal, persistent, receiver-effective relation with the ongoing distributed state. Local decoding without entry, entry without immediate overt action, action without reportable entry, and report loss with preserved content are required adverse or dissociation cases. The *Conscious Boundary of Choice* paper retains the complete entry/action/exit topology; this manuscript uses the boundary to keep the constitutive Qualia Tomogram claim falsifiable.

No-report binocular-rivalry work and experiments separating gamma/P3 activity from visual awareness demonstrate why introspection, action, and post-perceptual processing must not be silently folded into the conscious-content endpoint [56,57].

27 Physical Route Separation

Transmembrane and synaptic currents contribute to local extracellular electric fields. Local field geometry can in some preparations alter cellular excitability or timing. That establishes a candidate physical route, not one global display and not the complete tomogram. The model therefore keeps chemical synapses, electrical synapses, common input, volume transmission, neuromodulation, and local ephaptic influence as separate or interacting routes. Connectome harmonics and neural-field eigenmodes show that structural geometry can constrain distributed modes [38,39]. Slice, cortical, and cell-class-specific experiments show that endogenous or applied fields can modulate activity under bounded conditions [40–43]. Neither result establishes that one eigenmode or extracellular field is the whole experienced scene.

For an extracellular-field term to remain in the mechanism, it must add held-out predictive value after synaptic, common-input, movement, arousal, filtering, reference, and measurement-spread controls. A route-specific perturbation must change the prepared receiver in the predicted direction, and matched rescue must restore the relation rather than merely raise global activity. Failure narrows the physical route without erasing the broader receiver-relative reconstruction hypothesis.

28 Application, Mathematics, And Retirement Gates For The Successor

Drafts 2–10 will convert the present source-bound reconstruction into evidence:

- a source and primary-literature atlas with exact claim bindings;
- an executable generalized-reconstruction application and frozen benchmark;
- the preserved and, if feasible, executed TCNR motor-cortex protocol;
- machine-readable raw results and figures regenerated from those results;
- system-context and component-architecture diagrams;
- numerical recovery, null-space, perturbation, rescue, drift, and ablation tests;
- a mathematical specification with symbol and assumption tables;
- machine-checked finite-model lemmas for identifiability and consistency boundaries;

- an implementation-to-equation and implementation-to-proof map; and
- a losslessness and strongest-form adversarial audit.

The Qualia Tomogram center is retired or narrowed if capacity-matched alternatives predict every held-out view, perturbation, and action equally well without the declared projection structure; claimed views are not conditionally distinct after nuisance control; route-specific perturbation yields no selective error; rescue works only through nonspecific activation; a single reconstruction score maps indiscriminately to different contents; Fourier-family terminology adds no predictive constraint; or the receiver-relative variables cannot be measured reliably. A successful application proves only the behavior of its declared implementation. A successful formal proof establishes only its explicit assumptions and conclusion. Biological and phenomenal claims still require biological and consciousness-relevant evidence.

29 Biological Forward Operators: From Learned Receiver To Active Sampling

The generalized inverse problem becomes a biological hypothesis only when the forward operator is more than an unnamed nonlinear box. A neuron receives events at different dendritic branches while its membrane, channels, receptors, inhibition, intracellular state, recent activity, and larger brain-body context are already changing. The same nominal source event can therefore produce different outputs in two receivers, or in the same receiver at two times. The paper’s reconstruction must predict those conditional differences; otherwise the biological vocabulary adds no constraint.

This premise has a substantial established foundation. Thin dendrites can behave as nonlinear computational subunits; dendrites can discriminate the direction and velocity of temporal input sequences; and in-vivo work shows local dendritic organization and nonlinear events contributing to sensory selectivity [63–68]. Learning can alter spine morphology, AMPA-receptor availability, intrinsic dendritic excitability, and branch-specific plasticity [69–74,84]. Those results do not establish the SAN synthesis. They establish that receiver structure and state cannot safely be discarded from the forward model.

29.1 The receiver state is part of the measurement model

Let the declared state of receiver (i) be

$$\Theta_i(t) = (M_i, W_i(t), R_i(t), G_i(t), X_i(t), H_i(t), Z_i(t)), \quad C_t = (b_t, a_{t-1}, u_t), \quad (23)$$

where (M_i) is branch morphology; (W_i) is synapse location and efficacy; (R_i) is receptor availability and state; (G_i) is the declared set of intrinsic conductances; (X_i) contains measured intracellular variables; (H_i) is recent activity history; and (Z_i) contains local inhibitory and neuromodulatory conditions. The context (C_t) separates body state (b_t) , the preceding action (a_{t-1}) , and measured task or environmental context (u_t) . These variables are neither assumed independent nor assumed observable at every scale.

Equation 16 is then expanded, not replaced, by

$$y_i(t) = F_i(q_t, o_i(t), a_{t-1}; \Theta_i(t), C_t) + \varepsilon_i(t), \quad (24)$$

where (q_t) is the candidate latent body-world state and $(o_i(t))$ is the partial input physically available to

receiver (i). A model that uses only (q_t) , or only a scalar rate, is a comparator. The biological forward-operator claim earns support only when measured components of $(Theta_i)$ improve held-out prediction after capacity, input energy, history, movement, arousal, and recording quality are matched.

The source genealogy is specific. An internally dated September 2012 note proposed neuronal topography as a learned program-like constraint [86]. A companion note distinguished connection from influence and treated a transmitted potential difference as reflecting receiver history and structural change [87]. The later publicly Git-fixed 2022 broccoli/red account described learned multimodal properties being detected at dendrites, transformed through cells and arrays, and re-expressed in a recurrent sensory-action loop [88]. The 2012 sources are ancestors of the receiver-state idea; they are not retroactively renamed PWD or NAPOT.

29.2 Branch computation, learned selectivity, and typed output

For a branch (b), a minimal testable abstraction is

$$d_{ib}(t) = \phi_{ib} \left[\sum_j w_{ijb}^+(t) (K_{ijb} * s_j)(t - \tau_{ijb}) - \sum_k w_{ikb}^-(t) (L_{ikb} * r_k)(t - \delta_{ikb}) + \mu_{ib}(C_t); \Theta_{ib}(t) \right], \quad (25)$$

where excitation and inhibition have separate measured kernels, delays, locations, and signs. The function (phi_{ib}) may contain a branch nonlinearity, but its capacity must be reported. The cell then transforms the branch states into a typed event,

$$x_i(t) = \Psi_i \left(\{d_{ib}(t)\}_{b=1}^{B_i}, V_i(t), H_i(t); \Theta_i(t) \right), \quad (26)$$

whose measured coordinates may include event time, phase relative to a declared reference, duration, waveform, burst structure, amplitude, and release consequence. No coordinate is presumed informative merely because it is measurable. Each must add held-out receiver or behavioral prediction under its own reliability and nuisance controls.

The morphology is not imagined to travel down the axon as a literal miniature tree. Learned morphology and cellular state parameterize the transformation. The outgoing event is a compressed, typed consequence of that transformation. This is the source-faithful repair of the early "shape transmission" language and the biologically testable bridge to PWD.

Plasticity makes the operator time-dependent:

$$\Theta_i(t+1) = \Pi_{\mathcal{A}_i} [\Theta_i(t) + \eta_i U_i(x_i(t), o_i(t), m_t, e_t, a_t)], \quad (27)$$

where (m_t) denotes declared modulatory signals, (e_t) an outcome or mismatch variable, and $(Pi_{\mathcal{A}_i})$ constrains the update to a biologically or computationally admissible set. Equation 27 does not assert one universal plasticity rule. It exposes the variables that a proposed rule must measure and the possibility that long-term use changes the operator itself.

29.3 Inhibition selects transformations and routes

Inhibition is not represented here as generic failure, noise, or destructive interference. Depending on cell type, location, conductance, and timing, it can subtract, shunt, delay, veto a local regenerative event, alter gain, or select which branch and downstream route can affect output. Tunable dendritic inhibition has been shown to regulate the transformation from synaptic input to action-potential output [74]. The model must therefore test branch- and route-specific inhibitory terms against simpler global-negative-gain controls.

Retinal direction selectivity provides an existence proof for the logic, not a universal-cortex shortcut. Individual starburst-amacrine dendrites can exhibit direction-selective calcium signals; specific inhibitory wiring helps determine preferred and null directions in downstream ganglion cells [75–77]. This supports the possibility that morphology, local dynamics, and inhibition jointly select a transformation. It does not show that every brain region implements one retinotopic algorithm or that a retinal computation is already NAPOT.

29.4 Projection is completed by the next receiver

An upstream array (A) does not send a finished scene into array (B). It produces a route-conditioned pattern that the receiving array transforms under its own state:

$$x_B(t) = F_B(P_{A \rightarrow B}(x_A(t); \rho_{A \rightarrow B}(t)), \Theta_B(t), C_t) + \epsilon_B(t). \quad (28)$$

The route ($ho_ \{A \rightarrow B\}$) declares conduction delays, synaptic targets, geometry, and any tested field or gap-junction term. Equation 28 makes three alternatives visible: a copied-message model, a generic recurrent model, and the SAN receiver-transform model. The SAN residual is supported only if receiver state and route structure improve held-out downstream prediction and if selective route perturbation and matched rescue behave as predicted.

Motor-to-sensory projections in the whisker system carry movement, touch, and object-location information, and touch responses can depend on the phase of active whisking [78,79]. In auditory cortex, a motor-cortical corollary-discharge pathway changes local dynamics before and during movement [80]. These are established examples of receiver context and returned motor information altering sensory processing. They do not by themselves establish one distributed tomogram.

29.5 Body state and action determine the next view

A passive input sequence is not always equivalent to an actively acquired one. Behavioral state can change visual responses [81], interoceptive tasks recruit systems that track internal body signals [82], and closed-loop stimulation during directed whisking can produce a task-specific illusory-touch report where the same stimulation outside the relevant active context does not [83]. Even the whisker itself can perform a morphological transformation whose variables depend on how the animal contacts the object [85]. The next observation is therefore modeled as

$$o_i(t+1) = S_i(q_{t+1}, b_{t+1}, a_t; \kappa_i(t)) + \xi_i(t), \quad b_{t+1} = B(b_t, a_t, e_t), \quad (29)$$

where (κ_i) is the physical sampling geometry. The decisive comparator replays the same nominal sensory sequence while removing control over acquisition. An active-sampling term survives only if it

predicts a held-out difference between active and matched replay conditions after movement, reward, arousal, and timing are controlled.

This closes the operation:

learned receiver state -> branch-specific excitation and inhibition -> typed cellular event ->

route-conditioned array projection -> receiving-array transformation -> action -> changed body-world

sampling -> plastic update of the next receiver state.

It also introduces the **Coadaptive Substrate Principle**: a long-lived read/write interface changes the biological or artificial substrate along the pathways it repeatedly uses. A decoder calibrated on day one and a stimulator calibrated from that decoder cannot be presumed valid later. The application must measure operator drift, recalibrate on held-out sessions, and test whether the same intended content can be recovered through changed carriers without producing off-target sensory, affective, action, or self-model effects.

29.6 The biological-operator analysis discriminators and failure conditions

The biological-operator analysis adds six mandatory experiment families to the application proof:

1. **Morphology/state gain:** measured receiver-state variables must improve held-out output prediction over input-, history-, and capacity-matched scalar and recurrent baselines.
1. **Branch intervention:** perturb a declared branch or conductance and predict the lost typed component; restore it under matched nonspecific drive and test selective recovery.
1. **Inhibitory routing:** compare branch-specific inhibition with a global negative-gain model and test whether the selected downstream route changes as predicted.
1. **Receiver swap:** present the same upstream event to differently tuned receivers and predict the interaction, not merely a fixed sender label.
1. **Active versus replayed sampling:** preserve the sensory sequence while removing control over sampling; require a preregistered context-dependent difference.
1. **Coadaptive read/write:** measure recalibration, carrier drift, off-target content, and self-model spillover across repeated closed-loop use.

The biological forward-operator layer is rejected or narrowed if its variables are unreliable, fail to add held-out value, are absorbed by a capacity-matched black-box comparator, do not survive receiver swap, or do not produce selective perturbation and rescue. Success would establish a better model of the declared data and intervention. It would not alone establish that the reconstructed state is a conscious experience.

30 Tonic Context, Informative Difference, And Typed Physical Routes

30.1 The problem before the terminology

Imagine looking at a stable red cup while moving the eyes and hand. Much of the nervous system is already active before any new contour, glint, mismatch, or motor correction arrives. Ongoing activity supplies the current context: which receptors are excitable, which pathways are open, which rhythms are coordinated, which predictions are active, and which bodily state is being regulated. If every event were interpreted in isolation, the system could not distinguish an expected continuation from a consequential change.

The computational problem is therefore relational. A receiver must compare what arrives with what its own recent state and route make probable. A recurrent, coordinated pattern can provide a stable reference, but coordination alone does not identify the color, contour, source, value, or action consequence. The content-bearing candidate is the structured difference that a particular receiver can detect against its current expectation and use to alter subsequent processing. Only after that ordinary problem is explicit do we use SAN's two compact names: the expected recurrent reference is the **tonic context**, and a typed, receiver-relative departure with a testable consequence is a **Phase Wave Differential (PWD)**.

Gamma research provides the historical reason for the distinction. Long-range and local synchronization can organize distributed activity, support binding, and predict communication between neuronal groups [89-95]. Yet gamma-band measurements also contain multiple phenomena, including broadband transients and microsaccadic contamination, and gamma can track task or post-perceptual processes without uniquely specifying awareness [57,99,100,107]. The unresolved issue is not whether synchronization matters. It is how a coordinated background can preserve or reveal the uncommon differences from which a receiver can discriminate one state, object, prediction error, or action route from another.

30.2 Conditional information against a tonic reference

Let $(X_{r,t})$ be a preregistered event description available to receiver (r) in analysis window (t), (H_{t-}) the permitted pre-event history, and (c_t) the task, body, and recording context. A tonic model is not a universal baseline or a fixed frequency band. It is a frozen predictive distribution for that receiver and context:

$$p_r(x \mid H_{t-}, c_t; \theta_r), \quad (30)$$

where (θ_r) is estimated without access to the held-out outcome. The conditional self-information of an observed event is

$$\mathcal{I}_r(x_t) = -\log_2 p_r(x_t \mid H_{t-}, c_t; \theta_r). \quad (31)$$

Across a declared distribution of windows, the corresponding conditional entropy rate is

$$h_r = \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T \mathcal{I}_r(X_{r,t}) \right]. \quad (32)$$

These equations apply Shannon's distinction between expected and unexpected events to a receiver-relative neural problem [96,97]. If phase locking is common under a frozen tonic model, its continuation can have low

surprisal even while consuming energy and coordinating many cells. A rarer excitation, inhibition, omitted event, delay, phase slip, or structured splay can have greater surprisal. This is a conditional statement, not a rule that synchrony is always uninformative or desynchronization is always informative. A rare artifact, seizure transient, or random perturbation can be highly surprising and biologically useless.

Low surprisal likewise does not mean low voltage, low metabolic cost, low causal importance, or subjective emptiness. Shannon information measures improbability under a specified model. Biological meaning requires the further receiver, route, and consequence tests below.

30.3 The typed PWD record

The measured departure must retain enough structure to say what changed. For sender (s), receiver (r), and window (w), define a typed departure vector

$$\Delta_{r \leftarrow s, w} = \left[\text{wrap}(\phi_s - \hat{\phi}_{r \leftarrow s}), \log \frac{A_s + \epsilon}{\hat{A}_{r \leftarrow s} + \epsilon}, \tau_s - \hat{\tau}_{r \leftarrow s}, q_s - \hat{q}_{r \leftarrow s}, \iota_s - \hat{\iota}_{r \leftarrow s} \right], \quad (33)$$

where phase, amplitude, arrival time, event quantity, and inhibitory or suppressive state are examples rather than compulsory universal coordinates. Hats denote values predicted from the frozen tonic model. A PWD is then recorded as

$$\text{PWD}_{r \leftarrow s, w} = (r, s, w, \mathcal{M}_r, \Delta_{r \leftarrow s, w}, \mathcal{I}_r, \rho, \Delta y_r, \Delta z), \quad (34)$$

with receiver model (*mathcal{M}_r*), candidate physical route (ρ), immediate receiver effect (*Delta y_r*), and later behavioral, physiological, or representational consequence (*Delta z*). This record is more specific than phase coding, a traveling wave, or a band-power change. Those established phenomena may supply coordinates or carriers; the PWD hypothesis concerns their receiver-relative difference from an expected relation and the consequence of that difference.

Excitation and inhibition can both create informative departures. An excitatory volley may advance or amplify a receiver; feedforward, feedback, or dendrite-targeting inhibition may delay, suppress, reroute, or prevent an expected response [74,101-104]. An omitted event can therefore be as typed as an added event. The sign, anatomical target, timescale, and receiving state must remain visible rather than being compressed into a generic claim that "more activity" carries more content.

30.4 Coefficient of variation and circular dispersion are coordinates, not synonyms

The coefficient of variation can quantify relative spread in a positive-valued window,

$$\text{CV}(x_{1:n}) = \frac{s(x_{1:n})}{|\bar{x}_{1:n}| + \epsilon}, \quad (35)$$

but it is not itself a PWD, Shannon information, neural meaning, or causal effect. It becomes unstable near a zero mean and discards ordering unless ordering is included elsewhere. Its value is operational: a preregistered CV feature can test whether the relative variability of amplitude, interval, conductance, or another positive measure contributes incremental held-out information.

Phase dispersion is a circular quantity. For phases (ϕ_i) and optional nonnegative weights (w_j),

$$R_w = \frac{\left| \sum_{j=1}^n w_j e^{i\phi_j} \right|}{\sum_{j=1}^n w_j}, \quad D_w = 1 - R_w. \quad (36)$$

High (R_w) indicates concentration; high (D_w) indicates dispersion. Neither statistic alone tells us whether a pattern is structured, receiver-readable, or useful. Stable phase-of-firing relations can add information beyond spike counts [96], whereas active decorrelation can increase representational capacity or separate competing inputs [98]. SAN's proposal is therefore a division of labor: coordination maintains a shared temporal reference and structured departures preserve discriminable differences.

30.5 Structured difference must beat both continuation and noise

A candidate event must be more than improbable. Let (p_0) be the frozen tonic-continuation model, (p_N) a capacity-matched artifact or unstructured-noise model, and (p_S) a structured-departure model. The two preregistered likelihood-ratio tests are

$$\Lambda_0(x) = \log \frac{p_S(x)}{p_0(x)}, \quad \Lambda_N(x) = \log \frac{p_S(x)}{p_N(x)}. \quad (37)$$

Both must generalize to held-out data. The first asks whether the event departs from expected continuation; the second asks whether its structure is better explained than matched random variation, filtering leakage, volume conduction, eye movement, power, or another declared nuisance. A high-surprisal event that fails (Λ_N) is a surprise, not evidence for a rendered content operator.

The resulting prediction is non-monotonic. Too little coordination prevents a receiver from defining or maintaining the relevant relation. Excessively global locking can collapse distinctions and reduce the number of independently usable states. Unstructured dispersion can also destroy communication. Useful cognition should occupy a task- and scale-dependent region in which enough coordination preserves the reference while enough structured difference remains available for discrimination. This is the reason gamma synchronization alone could not complete a theory of conscious content: the common canvas and the differences rendered upon it perform different operations.

30.6 Every proposed physical route must be typed

The same statistical relation can arise through different physical mechanisms. The application must label the candidate route before intervention: chemical synapse, electrical synapse, axonal spike timing, subthreshold membrane current, dendritic field interaction, local ephaptic coupling, neuromodulatory or endocrine state, metabolic support, or measurement-level mixture. A population LFP or EEG relation is not automatically a direct cell-to-cell field transmission. Gap-junction coupling, common synaptic drive, and local field effects require different perturbations and spatial falloff predictions [101,102].

For route (rho_k), let the immediate receiver effect be estimated against a route-removed or route-matched counterfactual and let (G_k) be the capacity-penalized held-out prediction gain:

$$E_{r,k} = \mathbb{E}[Y_r \mid do(\rho_k = 1), H, c] - \mathbb{E}[Y_r \mid do(\rho_k = 0), H, c], \quad G_k = \Delta LL_{\text{heldout},k} - \lambda \Delta \text{Complexity}_k. \quad (38)$$

A supported route requires a reliable signed ($E_{r,k}$), positive preregistered (G_k), a predicted lost component under selective perturbation, and selective recovery under matched rescue. Phase-targeted closed-loop intervention is already experimentally feasible in bounded neural systems [105,106]. That technology supplies an important application primitive: detect a declared phase state, perturb at a locked phase, replay the same stimulation open-loop, and ask whether only the phase-contingent intervention changes the predicted receiver and task variable.

30.7 Early SAN genealogy and the July 24, 2013 canvas formulation

The mature operator arose from a longer conceptual sequence. SAN notes publicly fixed in 2022 distinguish coupling from content-bearing difference, describe synchronized activity as a common context, and develop phasic excitation and inhibition as departures that change what a receiving array registers [109-112]. The newly recovered July 24, 2013 Facebook post supplies an earlier bounded ancestor [113]. In that post, sensory experience is described as cycling synchrony; learned oscillatory patterns can reactivate internally rendered vectors, sounds, and worlds; smaller oscillations combine into larger brainwaves; and these dynamics "knit the canvas of conscious experience."

That source supports five early atoms: cycling sensory synchrony, learned oscillatory pattern formation, internally rendered sensory content, multiscale oscillatory composition, and a conscious-canvas metaphor. It does not contain the later terms PWD, NAPOT, SAN Tomogram, conditional self-information, or the present projection-slice equations. The genealogy is therefore additive and explicit: the 2013 source establishes an early canvas-and-rendering ancestor; the 2022 sources name and operationalize receiver-relative difference; the present paper turns that sequence into discriminating statistical and causal tests. The Facebook date is the date displayed by the captured post and its direct object route; displayed-platform dating remains separately labeled from immutable third-party public-custody evidence.

30.8 The tonic-context analysis application tests and failure conditions

The tonic-context analysis adds seven mandatory tests to the application proof:

1. **Frozen tonic model:** fit the receiver- and context-conditioned reference on development data only; disclose every history, task, body-state, and recording variable used.
1. **Surprisal calibration:** test whether estimated event probabilities are calibrated on held-out data; do not interpret miscalibration as biological information.
1. **Structured-versus-noise test:** require the departure model to beat tonic continuation and at least one capacity-matched nuisance or noise model.
1. **Coordinate ablation:** remove phase, amplitude, timing, rate, inhibition, CV, and dispersion coordinates separately; report which typed component supplies incremental value.
1. **Receiver swap:** hold the upstream event as constant as possible while changing receiver state; predict the resulting change in surprisal, immediate response, and downstream consequence.
1. **Route perturbation and rescue:** perturb one declared physical route, predict the lost component, and restore it without increasing nonspecific drive.

1. **Coordination-differentiation sweep:** vary coordination while preserving power and event count where possible; test the predicted task-specific dynamic range rather than a monotonic "more gamma is better" rule.

The tonic-PWD layer is narrowed or rejected if its probability estimates fail calibration, its structured model does not beat matched noise, its features add no value over equally capable raw-signal models, the effect does not depend on receiver state, or perturbation and rescue fail to isolate the declared route. Passing these tests would establish a reproducible receiver-relative information-and-effect operator in the declared system. The later drafts must still test whether distributed operators reconstruct held-out views, scale into a SAN Tomogram, and bear any defensible relation to reported qualitative content.

31 Distributed Reconstruction, Content Identity, And Conscious Reinstatement

31.1 From isolated features to one usable scene

No receptor receives broccoli. Some receptors transform wavelength distributions; others respond to pressure, texture, movement, volatile molecules, dissolved chemicals, temperature, tissue state, or the position and force of the body. Different circuits preserve different partial relations. Yet a person can recognize one object, distinguish it from the plate and hand, anticipate its taste and resistance, decide whether to eat it, and revise that expectation after a bite.

The problem is not solved by listing all detectable features. A flexible decoder may identify the label "broccoli" from one correlate without explaining which properties belong together, where the object is relative to the body, which boundary separates it from its surroundings, which absent property can be reinstated from memory, or what action would generate discriminating evidence. A constructive account needs a content identity that survives partial observation while remaining vulnerable to specific errors.

SAN proposes that the relevant unit is a changing distributed relation. Multiple receivers transform different views; recurrent exchange constrains which features, boundaries, body coordinates, memories, and action possibilities remain mutually consistent; action obtains another view; and the returned result updates the relation. When that active relation satisfies the declared reconstruction, boundary, and causal tests, the paper calls it a **SAN Tomogram**. The name summarizes the tested receive-transform-project- recur operation. It is not a picture presented to another observer: the participating network is changed by the relation and thereby performs the observing operation.

31.2 A typed content identity and segmentation contract

Let a candidate content at time (t) be

$$\mathcal{C}_t = (z_t, \Pi_t, b_t, m_t, u_t, \mathcal{R}_t, \mathcal{A}_t), \quad (39)$$

where (z_t) is a latent body-world state, (Π_t) a partition into candidate entities and boundaries, (b_t) the body-relative pose and internal state, (m_t) a memory or expectation relation, (u_t) uncertainty, $(\mathcal{R}_t)$ the declared receiver-consequence graph, and $(\mathcal{A}_t)$ the currently available action set. This tuple prevents "content" from collapsing into a stimulus label, one band, one cell list, one confidence value, or one report.

Segmentation is operationally important because a useful reconstruction must preserve both union and separation. Features assigned to one candidate object should predict one another under viewpoint and action, while features across a true boundary should not be forced into the same state. For candidate partition (Π), define

$$\mathcal{L}_{\text{seg}}(\Pi) = \sum_{G \in \Pi} \sum_{i,j \in G} d_{\text{within}}(i,j) + \alpha \sum_{G \neq H} \max(0, m - d_{\text{between}}(G,H)), \quad (40)$$

with distances and margin fixed before the confirmatory test. Object-based modulation and border-ownership signals show that cortical processing can carry grouping and boundary information [135,136]. Those findings make segmentation biologically plausible without making the particular SAN partition objective inevitable.

For held-out receiver set (J), condition (c), and action sequence (a), the reconstruction burden is

$$\mathcal{L}_{\text{view}} = \sum_{j \in J} w_j \ell_j \left(o_{j,t}, \hat{F}_j(\hat{z}_t, \hat{b}_t, \hat{m}_t, a_{<t}, c_t) \right), \quad \Delta_{\text{view}} = \mathcal{L}_{\text{best baseline}} - \mathcal{L}_{\text{SAN}}. \quad (41)$$

The SAN-style model earns a reconstruction claim only when (Δ_{view}) is positive on untouched views and survives capacity reporting, calibration, modality deletion, temporal shift, movement regression, and an equally capable nonlinear comparator. A good training reconstruction or a decoded label is not enough.

Body registration is part of the content rather than an optional annotation. Rubber-hand and body-map experiments show that vision, touch, proprioception, motor imagery, and effector organization can be joined or dissociated [129,131]. Corollary discharge supplies a bounded route through which a planned movement changes the interpretation of later sensory evidence [130]. The model therefore predicts separate errors for object identity, object location, body ownership, agency, self-location, action possibility, and expected refference rather than treating all of them as one accuracy score.

31.3 Virtual engrams and identity across carrier drift

Memory introduces a second inverse problem. The present reconstruction can preserve a learned relation even when the exact participating cells or response amplitudes change. Longitudinal studies report both drifting representations and preparation-dependent stability [48-52,114,115,118,119]. Distributed engram and reinstatement work further shows that a memory-related event can involve multiple regions and that partial cues can recruit larger learned relations [58,59,116,117,133,134]. These results motivate a carrier/content distinction; they do not dictate which invariant carries identity.

Let (g) be a permitted carrier transformation such as a registered cell permutation, turnover map, session alignment, or global phase offset, and let (Ψ) be a preregistered relational representation. A candidate content equivalence is

$$\mathcal{C}_t \sim_{\varepsilon, \mathcal{T}} \mathcal{C}_{t'} \iff \sup_{g \in \mathcal{T}} d(\Psi(\mathcal{C}_t), \Psi(g \cdot \mathcal{C}_{t'})) \leq \varepsilon \wedge \Delta_{\text{consequence}} \leq \varepsilon_y, \quad (42)$$

where the transformation family ($\mathcal{T}$), metric, tolerances, and downstream consequence are frozen before the held-out session. This is not unrestricted post hoc alignment. If ($\mathcal{T}$) is arbitrarily flexible, every state can be made equivalent to every other and the identity claim becomes empty.

The drift test separately measures carrier change and relational preservation:

$$D_{\text{carrier}} = d_C(C_t, C_{t'}), \quad D_{\text{relation}} = d_R(\Psi_t, \Psi_{t'}), \quad S_{\text{drift}} = D_{\text{carrier}} - \beta D_{\text{relation}}. \quad (43)$$

A strong virtual-engram case has measured carrier change, low held-out relational error, stable content-specific consequences, and selective loss when the proposed invariant is disrupted. Stable carrier codes are an equally important counterexample: the theory predicts flexibility where the system uses it, not mandatory drift in every brain area, species, task, or timescale [119]. Flexible human assemblies and ongoing-storage models also remain live alternatives [120,121].

SAN calls the temporarily reinstated relational event a **virtual engram**. Virtual means dynamically instantiated, not immaterial. Durable morphology, synapses, receptor state, inhibition, and connectivity constrain the event; a current ensemble carries it; and the relational reconstruction supplies the candidate content identity. The complete Virtual Engrams paper develops this program in depth [139]. Here it performs one bounded job: it explains how a SAN Tomogram can reinstate learned content without requiring an unchanging list of carriers.

31.4 Reconstruction, conscious entry, and exit are different transitions

A content can be locally decodable, capable of priming, or available to memory without organizing present experience. Masking, rivalry, report-independent paradigms, and recurrent-processing studies show why representation, access, report, confidence, and action must remain separate endpoints [56,57,122-128,138]. The Qualia Tomogram therefore treats conscious entry as an additional relation, not an automatic reward for successful reconstruction.

For candidate reconstructed content (q_j) and the active distributed rendering (c), define a bounded recruitment score

$$B_j(t) = w_K K_j(t) + w_\Phi \Phi_j(t) + w_R R_j(t) + w_I I_j(t) - w_X X_j(t) - w_C C_j(t), \quad (44)$$

where (K_j) is reciprocal held-out predictive coupling, (Φ_j) task-specific phase compatibility, (R_j) recurrence or persistence, (I_j) receiver-relative information, (X_j) route-specific exclusion or inhibition, and (C_j) a confound score for report, confidence, movement, and nonspecific state. The terms are measured separately before combination; no weight may be chosen on the confirmatory outcome.

Entry and exit use distinct thresholds and dwell times:

$$\text{Entry}_j(t) = \mathbf{1}[B_j(t) > \theta_{\text{on}} \text{ for } T_{\text{on}}], \quad \text{Exit}_j(t) = \mathbf{1}[B_j(t) < \theta_{\text{off}} \text{ for } T_{\text{off}}], \quad \theta_{\text{off}} < \theta_{\text{on}}. \quad (45)$$

The hysteresis makes a substantive prediction: a content can remain active briefly after the initiating cue weakens, and it can leave present experience while its learned substrate and some latent influence remain. The complete Conscious Boundary of Choice paper develops entry, exit, and route selection as a separate program [140]. In this paper, the bounded use is to prevent reconstruction success from being mislabeled as consciousness evidence.

The decisive causal dissociation is

$$\Delta_{\text{entry}} = \mathbb{E}[Y_{\text{content}} \mid \text{do}(B_j = B_j^+)] - \mathbb{E}[Y_{\text{content}} \mid \text{do}(B_j = B_j^-)], \quad |\Delta_{\text{local decode}}| \leq \delta, \quad (46)$$

where the intervention changes the boundary relation while preserving early local decodability within a preregistered equivalence margin. A matched rescue must selectively restore the same content endpoint, not merely raise global power, arousal, confidence, or report rate.

31.5 Action is a measurement operation

A reconstruction becomes useful by changing what the organism samples next. Looking, reaching, biting, or withholding action can each expose a different view. The policy should therefore be evaluated by how much the returned observation reduces uncertainty about the candidate content and improves a declared consequence:

$$a_t^* = \arg \max_{a \in \mathcal{A}_t} [I(Z_{t+1}; O_{t+1} \mid H_t, a) - \lambda \text{Cost}(a) - \mu \text{Risk}(a)], \quad q_{t+1} = U(q_t, o_{t+1}, a_t, b_{t+1}). \quad (47)$$

The return closes the content test. If a model predicts "broccoli" but cannot predict which touch, weight, smell, taste, or visual change follows a declared action better than matched alternatives, its label may be correlationally useful but its reconstruction claim is weak. The full embodied-rendering and coadaptive-interface papers develop this action/read/write grammar in greater detail [141,142]. Here it is the test that turns a static latent into a body-world estimate exposed to correction.

31.6 The broccoli read/write laboratory

The reference laboratory factorizes the target instead of writing one undifferentiated token. Let

$$z_{\text{broccoli}} = [z_{\text{color}}, z_{\text{form}}, z_{\text{texture}}, z_{\text{odor}}, z_{\text{taste}}, z_{\text{body}}, z_{\text{affordance}}, z_{\text{context}}]. \quad (48)$$

Each component is generated through a different frozen forward operator and is evaluated on withheld views. The laboratory then performs four classes of trial:

1. **Reconstruction:** infer the property bundle from incomplete, noisy, and conflicting views.
2. **Deletion:** remove one property or route and predict the specific missing relation and compensatory uncertainty.
1. **Insertion:** perturb one bounded component and test whether the intended content changes without uncontrolled spread into other sensory, affective, action, or self-model dimensions.
1. **Drift and rescue:** change carrier identity or operator parameters, test whether the relation remains recoverable, disrupt the declared invariant, and restore it with an exact rather than nonspecific rescue.

This design directly addresses a risk suggested by feature overdrive in artificial and biological systems. A successful intervention should make a property available in the intended relation. It should not force every downstream interpretation toward that feature or destabilize the person's body and identity model. The possibility that an inserted broccoli-like pattern produces a broader "broccoli self" hallucination is therefore not a joke or rhetorical flourish; it is a preregistered self-model spillover endpoint.

For target component (k), define specificity and spillover as

$$\text{Spec}_k = \frac{|\Delta y_k|}{\epsilon + \sum_{j \neq k} \omega_j |\Delta y_j|}, \quad \text{Spill}_{\text{self}} = d\left(s_{\text{self}}^{\text{post}}, s_{\text{self}}^{\text{pre}}\right), \quad \text{Spill}_{\text{off}} = \sum_{j \notin \mathcal{T}_k} \omega_j |\Delta y_j|. \quad (49)$$

A read result requires held-out content prediction. A write result additionally requires target-specific change, dose and timing control, route localization, tolerable off-target effects, and matched rescue. A phenomenal claim further requires convergent content reports or behavior, objective discrimination, body/self endpoints, and dissociation from demand, confidence, arousal, motor preparation, and generic salience. These gates preserve the long-range NerveGear destination while keeping the first application scientifically bounded.

31.7 The distributed-content analysis model contest and failure conditions

The distributed-content analysis adds eight mandatory comparisons:

1. label decoder versus full typed content identity;
2. feature bag versus boundary-preserving segmentation;
3. fixed-carrier model versus preregistered relational invariance;
4. stable-subspace, latent-alignment, and ongoing-storage drift alternatives;
5. recurrence-only versus typed view-prediction reconstruction;
6. local decodability versus conscious-entry and exit endpoints;
7. passive replay versus action-selected returned evidence; and
8. target insertion versus nonspecific activation, global feature overdrive, and self-model spillover.

The distributed-content layer is narrowed if a label-only, feature-bag, fixed-carrier, generic recurrent, or capacity-matched latent model predicts the same held-out views and consequences. The virtual-engram claim is narrowed if the declared invariant fails across drift or if a simpler ongoing-storage model explains the data. The conscious-entry claim is narrowed if the boundary relation adds nothing beyond attention, confidence, report demand, arousal, motor preparation, and local signal strength. The read/write claim fails if decoding cannot be converted into selective intervention, or if apparent rescue is explained by global activation. A passing synthetic laboratory establishes that these distinctions can be implemented and can fail. Biological and phenomenal evidence require their own experiments.

32 Executable Reference Applications And Measured Synthetic Results

32.1 The problem before the prototype

Sections 20–30 specify a reconstruction architecture, but a specification can hide contradictions. A paper can name receivers, forward operators, withheld views, phase relations, interventions, and rescue without showing that all of those pieces can coexist in one running system. The immediate engineering problem is therefore narrower than the biological claim: can the declared operations be instantiated, tested against strong alternatives, made to fail selectively, and reproduced from raw code and a fixed seed?

Two deterministic applications answer that bounded question. The first is a generalized multi-view inverse problem. The second is a carrier-drift and read/write laboratory built around the broccoli example. Both are synthetic by design. Their purpose is to expose the paper’s computational commitments before biological data are used, not to substitute synthetic success for neural evidence.

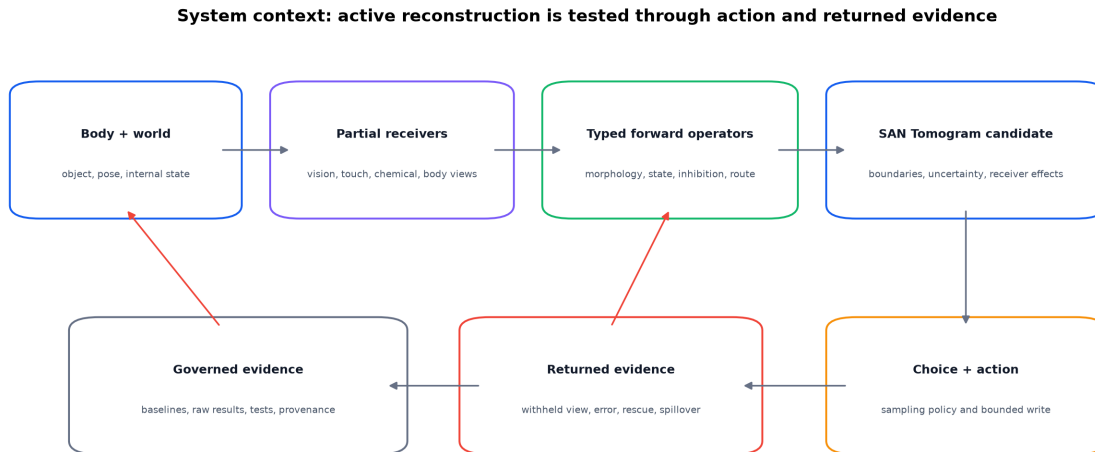


Figure 1: Figure 1. System context. Partial receivers feed typed forward operators; the candidate reconstruction guides a bounded action; returned evidence tests the reconstruction; and the complete record enters a governed evidence layer. The feedback arrow is an evaluation route, not an assumption that the estimate is correct.

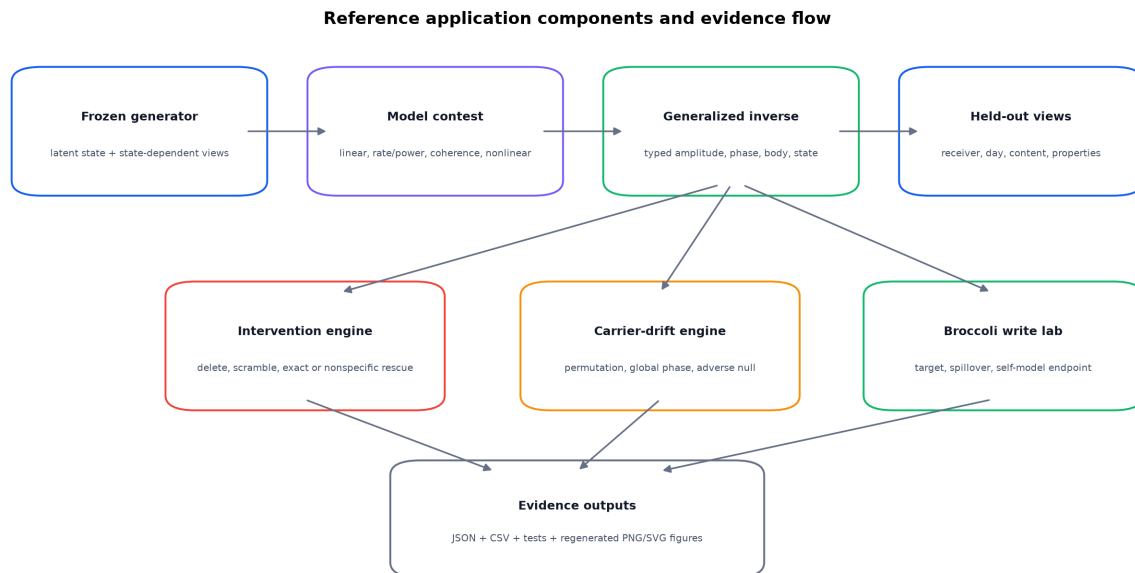


Figure 2: Figure 2. Component architecture. The generator, inverse models, baseline contest, perturbation/rescue engine, read/write laboratory, metrics, tests, and artifact manifest are separate components. This separation prevents a successful decoder from silently becoming a successful intervention or consciousness claim.

32.2 Generator, held-out views, and inverse objective

The primary generator uses four content families—broccoli, steak, apple, and cup—and an eight-coordinate latent state for color, form, texture, odor, taste, body relation, affordance, and context. Each noisy sample is transformed through five receivers with seven channels per receiver. Amplitude and circular phase depend on content, body state, context, day, and receiver state. For receiver (i), channel (k), sample (r), and day (d), the implemented forward map is

$$\begin{aligned} A_{rik}^{(d)} &= W_{ik}^{(d)} z_r + 0.11 (V_{ik} z_r)^2 + U_{ik} b_r + s_{ri} q_{ik} + c_r g_{ik} + \varepsilon_{rik}^A, \\ \theta_{rik}^{(d)} &= P_{ik}^{(d)} z_r + U_{ik}^\phi b_r + s_{ri} q_{ik}^\phi + c_r g_{ik}^\phi + \varepsilon_{rik}^\phi, \\ Y_{rik}^\phi &= \left(\sin \theta_{rik}^{(d)}, \cos \theta_{rik}^{(d)} \right). \end{aligned} \quad (50)$$

Here $(W^{(d)})$ and $(P^{(d)})$ drift with day, while the observed body, context, and receiver-state terms make the transformation explicitly receiver-conditioned. The primary split contains 900 training samples from days 0 and 1 and 320 untouched samples from day 2. Receivers 0–3 are available to the inverse; receiver 4 is reserved for hidden-view prediction.

The generalized inverse begins from a linear estimate and minimizes a declared amplitude, circular-phase, and regularization objective by deterministic gradient steps:

$$\begin{aligned} \hat{z}_r = \arg \min_z & \left[\sum_{i \in \mathcal{O}} \sum_k \left(A_{rik} - \hat{A}_{ik}(z, b_r, c_r, s_{ri}, d_r) \right)^2 \right. \\ & + \lambda_\phi \sum_{i \in \mathcal{O}} \sum_k \left\| Y_{rik}^\phi - \hat{Y}_{ik}^\phi(z, b_r, c_r, s_{ri}, d_r) \right\|_2^2 \\ & \left. + \lambda_z \|z\|_2^2 \right]. \end{aligned} \quad (51)$$

The primary endpoints are latent reconstruction error and prediction error on receiver 4:

$$\text{RMSE}_z = \sqrt{\frac{1}{ND} \sum_{r=1}^N \|z_r - \hat{z}_r\|_2^2}, \quad \text{RMSE}_{\text{view}} = \sqrt{\frac{1}{M} \sum_{m=1}^M (y_m^{\text{held}} - \hat{y}_m^{\text{held}})^2}. \quad (52)$$

32.3 Baseline contest

The full inverse is compared with scalar-coherence and rate/power summaries, a raw linear ridge decoder, a capacity-matched nonlinear random-feature decoder, a linearized tomography inverse, and generalized inverses without receiver state or phase. The capacity-matched control uses 1,624 declared coefficients, within 3.4% of the full operator’s 1,680. An earlier implementation accidentally allocated 21,218 coefficients to that control; the defect was found, corrected, and excluded from the recorded result.

Model	Latent RMSE	Held-out-view RMSE	Content accuracy	Property F1
Scalar coherence ridge	0.099731	0.094127	0.834375	0.947517
Rate/power ridge	0.053081	0.054902	0.996875	0.973097
Raw linear ridge	0.014871	0.031531	1.000000	0.995384
Capacity-matched nonlinear control	0.121111	0.107325	0.865625	0.922815
Linear tomography	0.042874	0.043837	1.000000	0.976758
Generalized without receiver state	0.037917	0.044703	0.996875	0.981428
Generalized without phase	0.033003	0.039605	1.000000	0.981121
Generalized full	0.013853	0.031271	1.000000	0.992752

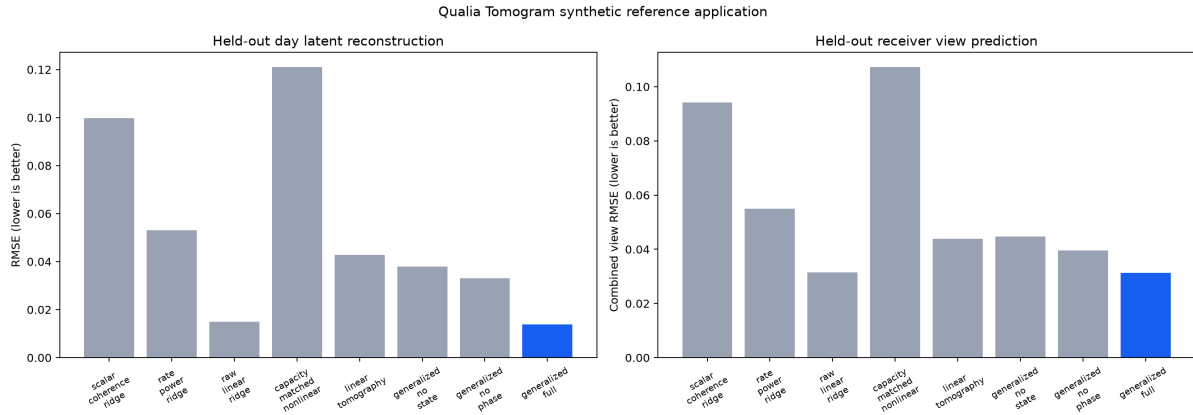


Figure 3: Figure 3. Primary model contest. The full typed inverse has the lowest error in the primary run and strongly separates from the mechanistic ablations. Content accuracy saturates for several models, showing why class labels alone are too weak to establish reconstruction.

32.4 Six-run replication and the raw-linear boundary

Six additional runs use independent deterministic samples, 600 training observations, 220 test observations, and 140 inverse iterations. Let $(e_{m,r})$ be the latent RMSE of model (m) in replication (r). The paired descriptive difference is

$$\bar{\Delta}_{m,\text{full}} = \frac{1}{R} \sum_{r=1}^R (e_{m,r} - e_{\text{full},r}), \quad R = 6. \quad (53)$$

Model	Mean latent RMSE	SD	Mean held-out-view RMSE	SD
Raw linear ridge	0.015190	0.000521	0.031156	0.000562
Linear tomography	0.045413	0.000856	0.045226	0.000876
Generalized without receiver state	0.038002	0.001693	0.044631	0.001303
Generalized without phase	0.035929	0.000617	0.040723	0.000746
Generalized full	0.015823	0.000412	0.031397	0.000720

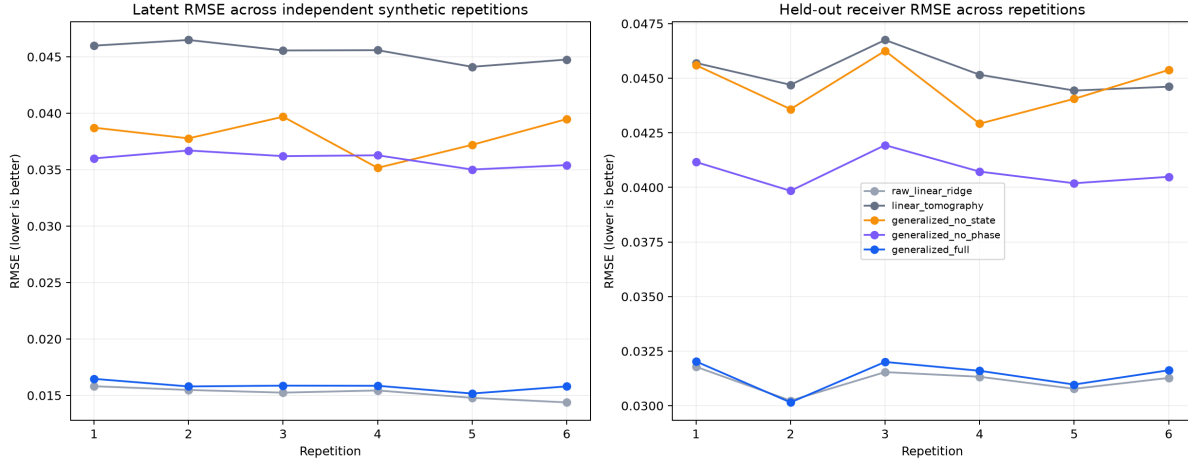


Figure 4: Figure 4. Replication stability. The full model retains a large advantage over linear tomography and the state- and phase-ablated versions. Raw linear ridge is slightly better on average. With only six synthetic replications, this is reported as a descriptive near-tie, not an inferential superiority claim.

This boundary matters. The generator was designed so that typed physical variables are available, but a flexible linear decoder also receives the raw amplitude, sine, cosine, body, context, day, and state inputs. The current application therefore shows that the typed inverse is sufficient, interpretable, and sensitive to its declared components. It does not show that its inductive bias is required whenever all raw variables and labels are available. A biological result must beat strong decoders under held-out receivers, days, conditions, and interventions—not merely recover the correct class.

32.5 Deletion, scramble, and rescue

The causal logic is implemented inside the synthetic model. Receiver 2 is removed; the exact receiver data are then restored; a norm-matched but sample-shuffled receiver is substituted; and the receiver’s phase relations are scrambled while amplitude is preserved. For an intervention (u), the error increment and rescue fraction are

$$\Delta E_u = E_u - E_{\text{intact}}, \quad \text{Rescue}(u) = 1 - \frac{E_u - E_{\text{intact}}}{E_{\text{deleted}} - E_{\text{intact}}}, \quad (54)$$

with the fraction interpreted only where the denominator and intervention are commensurate.

Condition	Latent RMSE	Content accuracy	Property F1
Intact	0.013853	1.000000	0.992752
Receiver deleted	0.017425	1.000000	0.991294
Exact rescue	0.013853	1.000000	0.992752
Matched nonspecific rescue	0.093700	0.946875	0.937554
Receiver phase scramble	0.073527	0.996875	0.955788

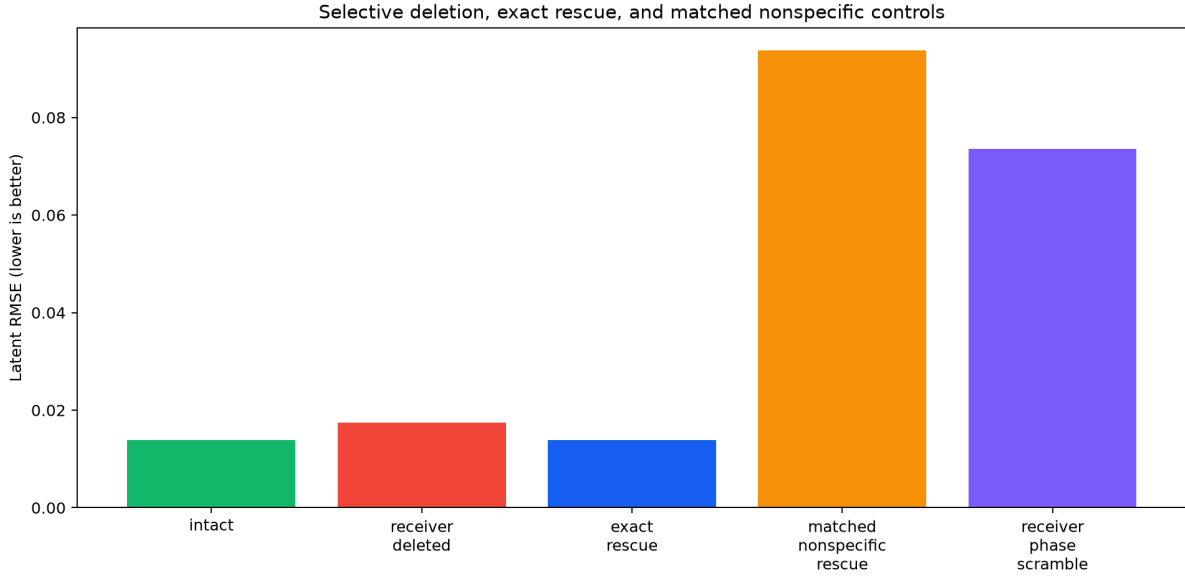


Figure 5: Figure 5. Perturbation and rescue. Exact replacement restores the intact result. Equal-scale but nonspecific replacement and phase scrambling do not. Because these manipulations occur in the generator, they establish causal dependence only for this synthetic implementation.

32.6 Carrier drift and an adverse phase-independent generator

The second application asks whether a content type can survive changed carrier identity. Twenty-four carrier channels are permuted and subjected to a global phase offset at test. Fixed-carrier, global-summary, amplitude-geometric, phase-relational, and combined relational features are compared by nearest-centroid classification. For representation (f), a bounded drift criterion is

$$\text{Inv}(f) = \text{Acc}(f(X_{\text{train}}), f(TX_{\text{test}})), \quad \text{PhaseGain}(f) = \text{Inv}_{\text{structured}}(f) - \text{Inv}_{\text{phase-independent}}(f). \quad (55)$$

Feature family	Structured drift accuracy	Adverse phase-independent accuracy
Fixed carrier	0.278125	0.300000
Global summary	0.990625	0.750000
Amplitude geometry	0.996875	0.993750
Phase relational	0.846875	0.237500
Complete relational	0.925000	0.678125

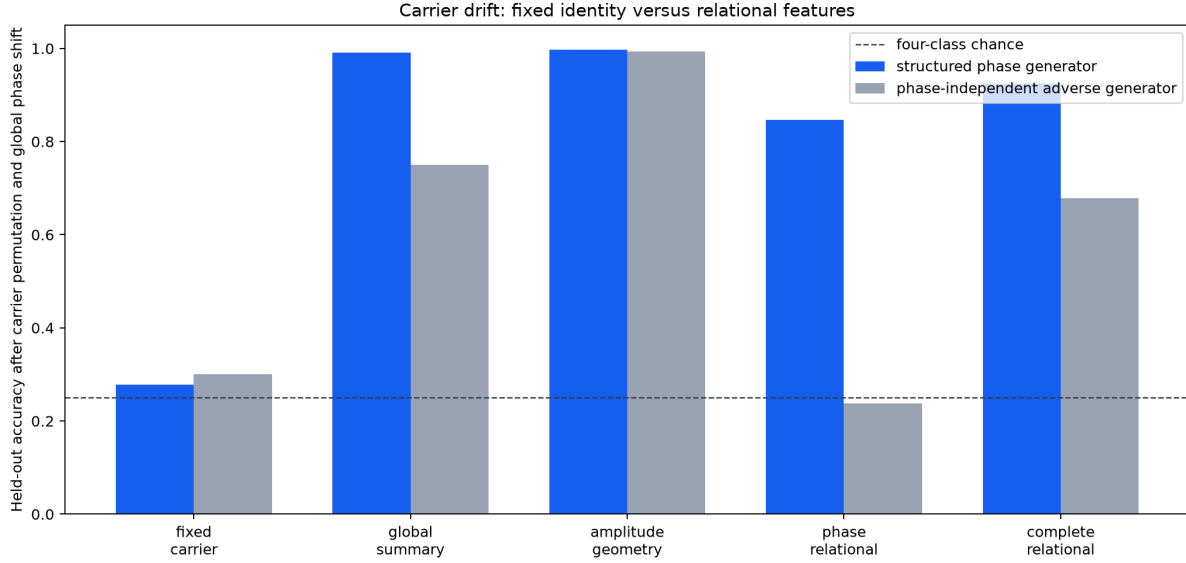


Figure 6: Figure 6. Carrier drift. Phase-relational features are useful when the generator carries content in phase relations and fall to approximately four-class chance when phase is randomized. Amplitude geometry is the strongest invariant in both conditions. The adverse generator is therefore a required negative result: the application does not manufacture phase necessity when the world does not contain it.

32.7 Bounded write, overdrive, and self-model spillover

The broccoli laboratory next separates one target coordinate—taste—from all off-target coordinates. A safe write adds a bounded taste change with small declared coupling. A matched nonspecific write has comparable norm but the wrong direction. Overdrive increases the target change while also perturbing context and body relation. Exact and nonspecific rescues begin from a broccoli state whose taste coordinate was deleted. The implemented metrics are

$$\text{Spec}_k = \frac{|\Delta z_k|}{\epsilon + \sum_{j \neq k} |\Delta z_j|}, \quad \text{Spill}_{\text{off}} = \sum_{j \neq k} |\Delta z_j|, \quad \text{Spill}_{\text{self}} = |\Delta z_{\text{body}}|. \quad (56)$$

Condition	Target effect	Off-target spill	Body/self-model spill	Specificity
Safe targeted write	0.520000	0.140400	0.031200	3.703704
Matched nonspecific write	0.043261	0.970288	0.177801	0.044586
Overdrive write	0.806000	2.079290	0.937620	0.387632
Exact rescue	0.730000	0.000000	0.000000	exact
Matched nonspecific rescue	0.052953	1.520439	0.022970	0.034827

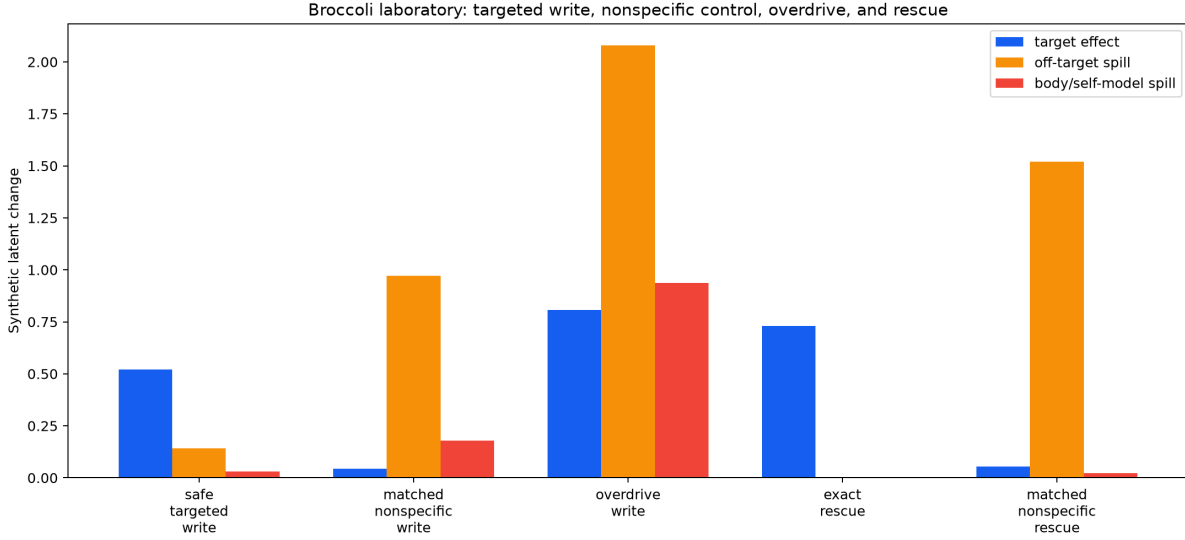


Figure 7: Figure 7. Read/write safety endpoints. A larger target effect is not automatically a better result. Overdrive produces substantially greater off-target and body/self-model change. Exact rescue returns the deleted vector with zero reconstruction error; matched nonspecific rescue has error 0.351513.

The term *self-model spillover* is operational here: it is change in the declared body-relation coordinate. It is not a simulated hallucination or subjective report. In a biological study the corresponding endpoint would require body-ownership, self-location, agency, perceptual, physiological, and safety measures rather than one latent coordinate.

32.8 Reproducibility and the application claim

The reference source, configuration, raw results, flattened tables, seven figures in PNG and SVG, tests, and hash manifest are stored with the manuscript. Ten automated checks pass. A second complete execution on the D: drive produces the same 23 artifact paths and SHA-256 hashes. The complete application claim is therefore

$$\mathcal{A}_{\text{pass}} = \mathcal{R}_{\text{heldout}} \wedge \mathcal{B}_{\text{explicit}} \wedge \mathcal{P}_{\text{selective}} \wedge \mathcal{Q}_{\text{rescue}} \wedge \mathcal{F}_{\text{adverse}} \wedge \mathcal{H}_{\text{reproducible}}, \quad (57)$$

where the terms denote held-out reconstruction, explicit baselines, selective perturbation, discriminating rescue, an adverse-generator failure condition, and deterministic reproducibility. This conjunction passes for the synthetic reference applications. It does not contain a biological-validity or consciousness term.

The next empirical gate is a preregistered neural-data application with frozen preprocessing, independent development and final-test subjects, strong raw-signal comparators, held-out views, route-specific perturbation, matched rescue, and report-independent controls. The next mathematical gate is to formalize the finite-dimensional projection-slice special case, the typed generalized operator, the identifiability assumptions, and the exact-rescue implication without claiming that the formal model proves its biological premises.

33 Identifiability, Invariance, And The Exact Mathematical Burden

33.1 One reconstructed vector can hide five different questions

A low reconstruction error does not answer whether the solution is unique, stable to noise, locally or globally identifiable, invariant under justified carrier changes, or semantically anchored to the content names used by the experimenter. The mathematical analysis separates those questions. Let the candidate state be $(z \in \mathcal{Z} \subseteq \mathbb{R}^D)$, receiver state be (s_i) , body and experimental context be $((b,c))$, and slow drift or day be (d) . A typed receiver is

$$F_i : \mathcal{Z} \times \mathcal{B} \times \mathcal{C} \times \mathcal{S}_i \times \mathcal{D} \longrightarrow \mathcal{Y}_i, \quad F_{\mathcal{O}}(z; \xi) = (F_i(z; \xi_i))_{i \in \mathcal{O}}. \quad (58)$$

The observed set $(\mathcal{O})$ supplies the inverse; a receiver outside that set supplies a hidden-view test. This typed form prevents nuisance variables, receiver state, and content coordinates from being exchanged silently.

33.2 Linear uniqueness and stability

After known offsets are removed, the linear special case is

$$y = Hz + \eta, \quad H \in \mathbb{R}^{M \times D}. \quad (59)$$

If $(\text{rank } H = D)$ and $(\eta = 0)$, then $(H^\top H)$ is positive definite and the least-squares solution is unique:

$$\hat{z} = (H^\top H)^{-1} H^\top y = H^\dagger y = z. \quad (60)$$

With observation error, the Moore–Penrose inverse gives

$$\|\hat{z} - z\|_2 \leq \|H^\dagger\|_2 \|\eta\|_2 = \frac{\|\eta\|_2}{\sigma_{\min}(H)}. \quad (61)$$

For a hidden linear receiver (G) ,

$$\|G\hat{z} - Gz\|_2 \leq \frac{\|G\|_2}{\sigma_{\min}(H)} \|\eta\|_2. \quad (62)$$

The result explains why more views can help without making the slogan “more views are always sufficient.” The frozen four-receiver day-2 matrix has 28 rows, 8 columns, rank 8, smallest singular value 1.250497, condition number 2.606653, and noise-free reconstruction error (1.03×10^{-15}) . In 256 random hidden-view perturbation checks, the largest observed output/input norm ratio is 1.998723, below the operator-norm bound 2.174408, with zero violations.

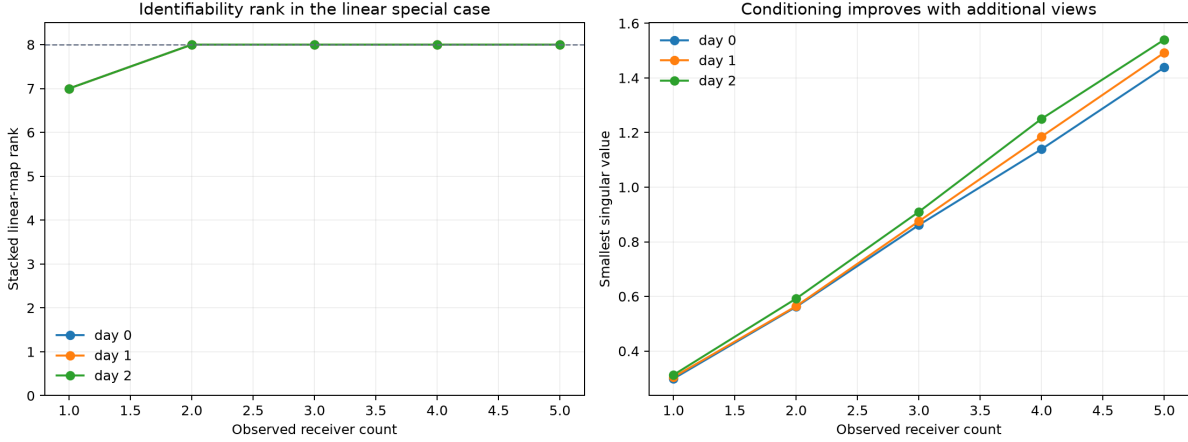


Figure 8: Figure 8. Rank and conditioning. One seven-channel receiver cannot span an eight-dimensional latent space. Two or more frozen receivers reach full rank in this generator, and the smallest singular value rises as additional views are added. This plot describes the frozen linear special case, not a universal law of neural recordings.

33.3 The null-space counterexample

If $(\text{rank } H < D)$, rank-nullity gives a nonzero $(v \in \ker H)$. Therefore

$$H(z + v) = Hz + Hv = Hz, \quad v \neq 0. \quad (63)$$

This is a constructive non-identifiability result: distinct candidate contents can be observationally identical under an insufficient view. In the frozen one-receiver case, rank is 7. A latent displacement of norm 0.75 changes the observation by only (3.52×10^{-16}) numerical error. No optimizer, larger language model, or more confident label can recover the missing dimension from that observation alone.

The appropriate responses are to add a genuinely independent view, impose a justified prior, select a new action that produces informative returned evidence, or narrow the claim. Calling the estimate a tomogram does not remove the null space.

33.4 Nonlinear local rank is not global identity

For a continuously differentiable stacked map, a full-column-rank Jacobian at (z_0) implies that some $(D \times D)$ coordinate minor is nonsingular. The inverse-function theorem then supplies a neighborhood in which those selected observations locally determine (z) :

$$\text{rank } J_{\mathcal{O}}(z_0) = D \quad \Rightarrow \quad F_{\mathcal{O}} \text{ is locally one-to-one on a sufficiently small neighborhood of } z_0. \quad (64)$$

The frozen 84-by-8 Jacobian has numerical rank 8 at broccoli, steak, apple, and cup, with smallest singular values 2.0676, 2.0657, 2.0711, and 2.0527. These checks are useful but blue: they do not exclude a distant alias, establish global injectivity, or show that biology uses the same coordinates.

33.5 The latent names require anchors

Let $(T : \mathcal{Z} \rightarrow \mathcal{Z}')$ be invertible, define $(z' = Tz)$, and replace every forward operator by $(F'_i = F_i \circ T^{-1})$. Then

$$F'_i(z') = F_i(T^{-1}Tz) = F_i(z). \quad (65)$$

The observation is unchanged even though the latent coordinate system changes. The executable orthogonal example moves the coordinate vector by norm 3.6214 while preserving the observation to (4.50×10^{-15}) numerical error. Consequently, reconstruction alone cannot earn semantic names such as *taste*, *body relation*, *affordance*, or *redness*. Those names must be anchored by controlled stimulus structure, cross-modal prediction, intervention, action consequence, or another independent criterion.

This is also why the paper’s content tuple is typed rather than merely numbered. A flexible latent may decode well while rotating together variables that the scientific claim needs to keep distinct.

33.6 Restricted carrier invariance

For amplitudes $(a_1, \dots, a_n)$ and phases $(\phi_1, \dots, \phi_n)$, define the sorted amplitude multiset and the sorted pairwise phase-relation multiset

$$\mathcal{A}(a) = \text{sort}\{a_i\}_{i=1}^n, \quad \mathcal{P}(\phi) = \text{sort}\{\cos(\phi_i - \phi_j) : i < j\}. \quad (66)$$

Both are invariant to a common carrier permutation. The phase relation is also invariant to a common phase offset because $((\phi_i + \alpha) - (\phi_j + \alpha) = \phi_i - \phi_j)$. The executable test applies a random carrier permutation and offset 1.731. Maximum differences are 0 for amplitude geometry, (8.74×10^{-16}) for phase relations, and 3.7837 for fixed-carrier features.

The restriction is load-bearing. Carrier loss, carrier-specific nonlinear phase warps, time-dependent relabeling, and unrestricted post-hoc transformations are not covered. A transformation family broad enough to align any two states would make “same content” unfalsifiable.

33.7 Exact rescue and selective writing

Let (D_k) delete coordinate (k) and let (Q_k) restore its exact original value. Componentwise equality gives $(Q_k(D_k(z), z_k) = z)$. Any deterministic readout therefore returns exactly to its original output. For a write (Δz) , the declared specificity remains

$$Q_k(D_k(z), z_k) = z, \quad \text{Spec}_k(\Delta z) = \frac{|\Delta z_k|}{\epsilon + \sum_{j \neq k} |\Delta z_j|}. \quad (67)$$

An isolated one-coordinate write has zero off-target spill by definition. This algebra does not imply that a neural stimulator can isolate the coordinate. The biological burden is precisely to show that the intended coordinate changes, the off-target coordinates do not, and exact rather than equal-energy nonspecific rescue restores the declared content and consequence.

33.8 Status map and next formal gate

The linear uniqueness, stability, null-space, reparameterization, restricted-invariance, exact-rescue, and isolated-write statements have explicit paper proofs. Under the proof-status contract they remain yellow until their exact finite or algebraic statements compile in a proof assistant, contain no admitted placeholders, and receive a theorem-scope audit. The frozen ranks, residuals, condition numbers, and Jacobian checks are blue. The mapping from mathematical objects to neurons, percepts, or consciousness remains gray.

Ten executable-mathematics tests pass, and a second D:-drive run reproduces all four generated mathematics artifacts byte-for-byte. The formal kernel will machine-check only the bounded algebraic kernel. It will not encode the biological premises as axioms and then announce that biology has been proved.

34 A Machine-Checked Kernel Without A Machine-Made Biology

34.1 What problem formal verification solves

A numerical reconstruction can look persuasive even when the mathematical statement behind it is vague. A proof assistant addresses one part of that problem: it forces every object, premise, and conclusion in a bounded theorem to be explicit and rejects a proof step that does not follow. It cannot determine whether a neuron is the declared object, whether the experiment measures the needed variable, or whether a latent coordinate is a qualitative experience. Those are empirical and interpretive questions.

The formal kernel therefore uses Lean as a verification instrument rather than as the application domain. The formal package contains fifteen theorems divided among reconstruction, restricted carrier invariance, and exact intervention algebra. The build completes under Lean 4.26.0 and Mathlib 4.26.0. A separate axiom audit completes successfully, and a source scan finds no standalone `sorry` or `admit` placeholder. Each green result below means only that the implication is proved under the stated assumptions.

34.2 A left inverse earns uniqueness; a decoder label does not

Suppose a forward map ($H : Z \rightarrow Y$) converts a candidate latent state into an observation and a reconstruction map ($L : Y \rightarrow Z$) is a left inverse of (H). The premise is not that (L) usually works or has low average error. It is the pointwise identity

$$(\forall z \in Z) L(H(z)) = z. \quad (68)$$

Lean theorem `leftInverse_exact` checks the identity's immediate reconstruction consequence. `leftInverse_observation_unique` checks that the same premise makes (H) injective: if two candidate states produce the same observation, they are equal. This is the clean abstract core of the The mathematical analysis uniqueness argument.

The premise is load-bearing. The kernel does not infer that a left inverse exists from an empirical decoder, a finite training score, or a matrix described as full rank. The mathematical analysis's matrix-rank and Moore–Penrose argument remains a yellow theorem at that greater level of specificity. The frozen rank calculation remains blue.

34.3 A missing direction creates a constructive ambiguity

The corresponding obstruction begins with an additive forward map and a nonzero latent displacement (v) that the map cannot see:

$$H(v) = 0, \quad v \neq 0 \quad \Rightarrow \quad H(z + v) = H(z), \quad z + v \neq z. \quad (69)$$

Lean separately checks the equal-observation step, the distinct-state step, and their conjunction in `nullDirection_sameObservation`, `nullDirection_distinctState`, and `nullDirection_nonidentifiability`. This is more than a warning that reconstruction may be difficult. It is a witness form: once the premises hold, two different states are observationally indistinguishable through that view.

The result also explains why naming a representation does not recover absent information. More optimization, confidence, or linguistic detail cannot choose between states that the observation map treats identically. The experiment must add an informative view or intervention, justify a prior, or narrow its conclusion.

34.4 Coordinates can change while observations stay fixed

Let $(T : Z \simeq Z')$ be an invertible change of latent coordinates. If the transformed forward map is defined by $(H' = H \circ T^{-1})$, then

$$H'(Tz) = H(T^{-1}Tz) = H(z). \quad (70)$$

`reparameterized_observation` machine-checks this identity. It is a formal reason not to confuse a recovered coordinate system with its scientific meaning. A model can preserve every observation while rotating or otherwise reparameterizing the latent variables. Labels such as *redness*, *taste*, *body relation*, and *affordance* therefore require independent anchors from stimulus control, cross-view prediction, intervention, action consequence, or another declared source of semantics.

34.5 Error bounds are conditional, not decorative

The full The mathematical analysis stability argument uses a singular value and a hidden linear operator. The green kernel checks a smaller implication that appears inside that argument. If an absolute error is already bounded and a scalar readout gain is nonnegative, multiplying by that gain preserves the bound:

$$|e| \leq B, \quad g \geq 0 \quad \Rightarrow \quad g|e| \leq gB. \quad (71)$$

This is theorem `bounded_scalar_readout`. It prevents a sign error in the propagated bound. It does not prove the preceding singular-value estimate, identify a biological gain, or establish a hidden-view operator norm. Those stronger claims remain yellow until their exact library-level statements are separately formalized.

34.6 The carrier-invariance family is deliberately small

A carrier transformation should be specified before the comparison, because an unrestricted transformation could align almost any two states. The kernel permits one shared relabeling (p) and one shared phase offset (α). With transformed phase ($\phi'(p(i)) = \phi(i) + \alpha$), corresponding pairwise differences satisfy

$$\phi'(p(i)) - \phi'(p(j)) = (\phi(i) + \alpha) - (\phi(j) + \alpha) = \phi(i) - \phi(j). \quad (72)$$

`relabel_at_image` verifies pointwise carrier correspondence. `pairwiseDifference_commonOffset` verifies cancellation of the common offset. `pairwiseDifference_relabelWithOffset` combines both operations, and `pairwiseReadout_relabelWithOffset` proves that any fixed readout of the preserved difference is also preserved. The last theorem covers cosine or another preregistered relation function without making that function part of the transformation chosen after seeing the result.

The package does not certify invariance to carrier deletion, carrier-specific phase warping, nonlinear time-dependent relabeling, or a post-hoc map selected to force agreement. The mathematical analysis's sorted amplitude and pairwise-relation collection claims remain yellow beyond the pointwise kernel formalized here.

34.7 Exact rescue and an isolated write are identities, not devices

The intervention kernel represents a state as a coordinate-indexed function. Deletion replaces coordinate (k) with zero; rescue writes its exact original value back; an isolated write updates only (k). The formal identities are

$$Q_k(D_k(z), z_k) = z, \quad W_k(z, a)_k = a, \quad j \neq k \Rightarrow W_k(z, a)_j = z_j. \quad (73)$$

`exact_delete_rescue` proves complete state restoration, and `exact_delete_rescue_readout` proves that every deterministic readout consequently returns to its original value. `isolated_write_target` proves the target coordinate takes the requested value, while `isolated_write_offTarget` proves every explicitly different coordinate remains unchanged.

These identities specify what an exact rescue or perfectly isolated write would mean. They do not show that an electrode, optical method, drug, ultrasound field, or other neural intervention can realize the operation. That empirical difficulty is not an inconvenience outside the theorem; it is one of the paper's central experimental burdens.

34.8 Proof-status readback

The machine-checked set is intentionally asymmetric. Fifteen compact implications are green. The full rank-to-pseudoinverse argument, singular-value/operator-norm bound, inverse-function local theorem, and collection-level sorted invariance remain yellow paper proofs. Frozen rank, conditioning, residual, and Jacobian results remain blue. Claims that the objects map onto neurons, rendered content, conscious access, or human stimulation remain gray.

This separation prevents two opposite errors. A formal proof is not dismissed merely because it is abstract: within its model, the checked implication is exact. But the proof is also not inflated into empirical

confirmation merely because a proof assistant accepted it. The formal kernel strengthens the chain of reasoning by making the algebra auditable while leaving every biological premise available for experimental failure.

35 Steelman-First Adversarial Audit And Decisive Failure Conditions

35.1 The objection must target the complete argument

A red team can appear rigorous while attacking a compressed substitute for the proposal. That is especially easy here because the manuscript contains phase, reconstruction, tomography, qualitative content, neural rendering, and consciousness in one lineage. The governing rule is therefore: before an objection counts, it must state the strongest actual claim, the observations on which it relies, the alternatives already tested, and the boundary the manuscript already accepts. Only then may it identify a missing premise, confound, or counterexample.

The strongest current claim is not that the synthetic application reads qualia. It is that one explicit receiver-conditioned multiview architecture can be executed, audited, compared with declared alternatives, and used to define sharper biological tests. The application and formal kernel establish internal consistency, reproducibility, and discriminating model behaviors. They do not establish the neural, phenomenal, or stimulation premises.

35.2 Evidence levels do not substitute for one another

Four levels must remain separate:

1. **Synthetic executability:** the operation runs in a declared artificial world.
2. **Neural prediction:** the locked representation adds held-out value in biological data.
3. **Neural causal specificity:** a targeted perturbation causes the predicted selective loss and matched rescue restores it.

1. **Phenomenal discrimination:** the effect distinguishes candidate conscious content from attention, report, confidence, arousal, movement, memory availability, and generic recurrence.

The non-implication is part of the method:

$$E_{\text{synthetic}} \not\Rightarrow E_{\text{neural}} \not\Rightarrow E_{\text{causal}} \not\Rightarrow E_{\text{phenomenal}}. \quad (74)$$

Evidence may support more than one level only when the corresponding measurements and controls are actually present. A deterministic replay belongs to Level 1. It is reproducibility within the recorded environment, not independent replication and not a biological result.

35.3 The generator may teach the model its own answer

The synthetic application is strongest as an existence and discrimination proof. It shows that the typed operator can be implemented, that ablations can fail, that a hidden view can be predicted, and that exact and nonspecific rescue can be distinguished. The serious objection is that its generator was designed with the

same conceptual vocabulary. A model can recover planted structure without that structure being the best description of a different world.

The existing capacity-matched controls, raw-signal baseline, six frozen replications, and adverse phase generator reduce but do not eliminate this risk. A stronger test freezes a family of generators unknown to the analysis author and evaluates the worst held-out advantage over the strongest comparator:

$$\Delta_{\text{gen}} = \min_{g \in \mathcal{G}_{\text{held}}} [L_g(M_{\text{best comparator}}) - L_g(M_{\text{QT}})] . \quad (75)$$

A positive value would support cross-generator predictive advantage under the declared loss. A nonpositive value would not erase executability; it would defeat the stronger generalization claim. This release must not describe this held-out-generator gate as passed, because it has not been run.

35.4 The strongest baseline is allowed to win

The full model beats its mechanistic ablations, but raw linear ridge is slightly better on average across the six smaller replications. That result is not a formatting nuisance. It means the current application does not establish general predictive superiority. The mechanistic model may still be valuable because it exposes receiver state, phase, deletion, rescue, and write variables that a raw predictor does not interpret in the same way. That is a claim about experimental articulation, not permission to ignore predictive loss.

The adverse generator is equally important. Phase-relational features succeed when phase structure is present and lose that advantage when it is absent; amplitude geometry is stronger in the present construction. SAN's proposal is therefore conditional: structured phase can be informative when receiver tuning and task physics make it so. The paper rejects the universal statement that phase coherence is always necessary or always the best content variable.

35.5 One dataset cannot answer a question it confounds

The proposed Brochier analysis is a legitimate protocol for predicting the cue-instructed grip condition. It cannot separate cue identity from grip instruction because the task couples them. No amount of cross-validation removes a design confound. A result must therefore be reported with that joint label, and an independently maintained motor-plan claim requires another experiment in which cue features and planned movement vary separately.

The same rule applies to conscious access. Report, attention, confidence, arousal, eye and body movement, memory availability, and generic recurrence are not interchangeable with experience. A candidate access result must survive matched no-report or orthogonal-report conditions and a preregistered perturbational endpoint before the paper may move from neural prediction to phenomenal discrimination.

35.6 Identifiability and semantics remain empirical burdens

Lean proves what follows if a left inverse exists; it does not prove that a recorded brain view has one. The rank-eight frozen matrix and local Jacobians are useful model checks, while the rank-seven counterexample shows why insufficient views fail. Global aliases, model misspecification, nuisance variation, and sampling remain possible.

Even a globally successful reconstruction would not name its own coordinates. The reparameterization theorem shows that observations can be preserved under a change of latent coordinates. Every content label therefore needs an independent anchor. A broccoli coordinate is not earned merely because the training set contained broccoli; its color, form, texture, chemical, body-relative, affective, and action consequences must be factorized and tested.

35.7 Invariance must be fixed before it succeeds

The current formal family—one common carrier relabeling and one common phase offset—is intentionally small. It gives a clean positive theorem. Biological drift may include carrier loss, new cells, nonlinear warping, changed sampling, and altered tuning, so the family may also be too narrow. Conversely, allowing an arbitrary alignment after seeing both states would make identity too easy to obtain.

The repair is procedural: preregister the transformation family, estimate it without the held-out content, apply it unchanged, and report out-of-family failures. Stable-carrier models remain required alternatives. “Neural drift” cannot serve as an automatic explanation whenever a preferred coordinate disappears.

35.8 Exact rescue defines the target but does not build the instrument

Exact restoration is valuable because it defines an unambiguous causal ideal. Its formal success is also partly tautological: if the exact coordinate is restored, the exact vector returns. The biological test must therefore measure selectivity rather than borrow it from notation. At minimum, a causal claim requires

$$\begin{aligned} \mathcal{C} = & \mathbf{1}[\text{selective loss after targeted perturbation}] \\ & \cdot \mathbf{1}[\text{exact or calibrated rescue beats sham}] \\ & \cdot \mathbf{1}[\text{rescue beats equal-energy nonspecific drive}] \\ & \cdot \mathbf{1}[\text{off-target spill remains below its bound}]. \end{aligned} \tag{76}$$

If any factor is zero, the complete selective-causality claim fails. A larger evoked response does not repair poor specificity. Sensory, affective, motor, autonomic, and self-model spill must be measured rather than assumed absent.

35.9 The title names a research object, not a completed measurement

The phrase *Qualia Tomogram* is retained because it names the proposed target: a distributed, receiver-conditioned reconstruction whose relations are hypothesized to participate in qualitative content. It does not mean that the current synthetic vector is a qualia measurement. Until neural prediction, causal specificity, and phenomenal discrimination are passed, the object remains a candidate SAN Tomogram and the paper remains a theory, methods, application-proof, and formal-specification paper.

This distinction also protects the no-inner-observer architecture. A successful tomogram would not be an image displayed to a second viewer inside the brain. It would be a causally active relation among receivers, body state, prediction, action, and returned evidence.

35.10 Historical evidence and scientific evidence have different jobs

The July 24, 2013 Facebook post strongly documents an early Micah-authored ancestor: cycling synchrony, learned oscillatory patterns, internally rendered vectors and worlds, multiscale combination, and a canvas of conscious experience. Its custody class remains a platform-displayed date in a screenshot frozen and hashed in 2026. It does not independently expose edit history, and it does not contain the later names or full operators.

Historical comparisons are governed in a separate source-audit program and do not enter the constructive neuroscientific, computational, or formal evidence reported here.

35.11 A conjunctive decision rule keeps partial success honest

The complete theory should not pass because one attractive endpoint succeeds while its alternatives or failure conditions are ignored. Let the decisive vector be

$$\mathbf{G} = (G_{\text{held view}}, G_{\text{strong baseline}}, G_{\text{specific perturbation}}, G_{\text{matched rescue}}, G_{\text{semantic anchor}}, G_{\text{phenomenal discriminator}}). \quad (77)$$

Each coordinate receives its own result. Synthetic executability presently passes. Deterministic replay and the bounded formal kernel pass within their declared domains. General cross-generator superiority, the biological neural endpoint, selective biological rescue, semantic saturation, and phenomenal discrimination remain open. A failure narrows the corresponding claim; it does not authorize deleting earlier evidence or promoting a different tier to replace it.

35.12 What survives the red team

The paper survives as a substantial research program because its core is operational: partial views, receiver-conditioned transformations, a shared estimate, held-out projection, explicit alternatives, perturbation, rescue, action, and returned evidence. It now has a working deterministic reference application, adverse cases, executable mathematics, a constructive non-identifiability witness, and a machine-checked algebraic kernel.

What does not survive is a stronger premature reading. The current package does not show that its typed model beats every raw predictor, that phase is universally decisive, that the Brochier task isolates motor planning, that local rank supplies global identity, that a latent name supplies its semantics, that exact rescue is technically available, or that a reconstructed neural state is conscious. Those are not concessions that weaken the method. They are the boundaries that make the positive results interpretable and the next tests decisive.

36 Data, Code, And Reproducibility

The companion archive contains the deterministic synthetic reference application, frozen configuration, source, tests, structured results, executable mathematical checks, conditioning data, model and intervention figures, Lean theorem sources, build and axiom receipts, theorem-status ledger, and adversarial register. Relative reproduction instructions and a SHA-256 manifest bind those materials to this manuscript version.

The prospective neural protocol uses the public Brochier et al. reach-to-grasp recordings identified in References 1 and 2; that third-party dataset is not redistributed with this paper. The manuscript locks its primary endpoint, comparator families, exclusions, perturbation logic, and failure conditions before a future neural execution. Synthetic seeds, model coefficients, and artifact-generation rules are supplied for the completed reference applications.

These materials establish reproducibility only within the declared synthetic, mathematical, and formal models. They do not constitute a completed neural analysis, biological intervention, consciousness result, or measurement of qualia.

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